



## Measuring the Gap Between Declared and Real Urgency in a Supply Chain Digital Twin

*A decision-support layer for the Smart Green Supply Chain programme*

ALLEN Sud-Ouest | Direction de l'Innovation  
Lab de Toulouse

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Promotion 2026 | Cycle Ingénieur 3<sup>e</sup> année  
Parcours double diplôme Cranfield University (DED)

Période du 13 avril 2026 au 16 octobre 2026  
3<sup>e</sup> année | Mission de fin d'études

**Rapport confidentiel : OUI**

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### Thématique du travail réalisé

Informatique | Génie industriel

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## Acknowledgements

My thanks go first to Sébastien Perthuisot, Research Programme Director, Quality & Supply Chain Expert at the Direction de l'Innovation, who set the direction of this work and then left me room to take it somewhere. The weekly review he held with me was where most of the difficult questions in this report were first asked.

I thank Émilienne Lardy and Selma Ferhat for their scientific supervision, and for the scrum and planning cadence that gave the six months their rhythm. Their reviews repeatedly sent me back to check a claim I had been ready to make.

I thank Liam Jones, who worked on the network representation side of the same programme. Sharing a weekly demo with someone solving a neighbouring problem did more for the quality of my own work than any amount of solitary effort would have.

I thank Dr David Gomez of ESTIA for the academic supervision and framing of this mission, and Dr Peter Sherar of Cranfield University for supervising the research strand that ran alongside it.

Finally, I thank the whole Direction de l'Innovation team at the Toulouse Lab for the welcome, and my family for their support across the whole of this engineering degree.

## Glossary

|                          |                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adequacy score</b>    | Agreement between declared and computed urgency at a node, scored on $[0, 100]$ with an asymmetric penalty that prices hidden risk at roughly twice false urgency.                                                                                                                                                                                                                   |
| <b>AHP</b>               | Analytic Hierarchy Process. Elicits criteria weights from pairwise comparisons and validates them through a consistency ratio.                                                                                                                                                                                                                                                       |
| <b>AR(1)</b>             | First-order autoregressive process. Used here to extrapolate a node's computed urgency series forward.                                                                                                                                                                                                                                                                               |
| <b>AUC</b>               | Area under the ROC curve. The probability that a randomly chosen positive case is ranked above a randomly chosen negative one. 0.5 is chance.                                                                                                                                                                                                                                        |
| <b>Brier skill</b>       | A probabilistic forecast's mean squared error, expressed relative to a reference forecaster. Positive beats the reference; negative is worse than it.                                                                                                                                                                                                                                |
| <b>Consistency ratio</b> | AHP's internal check on a set of pairwise comparisons. A questionnaire whose ratio exceeds the accepted threshold is refused.                                                                                                                                                                                                                                                        |
| <b>Declared urgency</b>  | The urgency an operator states for a node, elicited through the questionnaire. Written $U_d$ .                                                                                                                                                                                                                                                                                       |
| <b>DIN</b>               | Direction de l'Innovation. The entity responsible for ALTEN's own research and development, as distinct from client work.                                                                                                                                                                                                                                                            |
| <b>DISCO</b>             | The research programme's serious game and digital twin, simulating a satellite supply chain across 22 component suppliers.                                                                                                                                                                                                                                                           |
| <b>False urgency</b>     | The part of the gap where declared urgency exceeds computed urgency. Consumes shared capacity another flow needed.                                                                                                                                                                                                                                                                   |
| <b>GLEC Framework</b>    | Global Logistics Emissions Council methodology for calculating logistics emissions, aligned with ISO 14083.                                                                                                                                                                                                                                                                          |
| <b>Hidden risk</b>       | The part of the gap where computed urgency exceeds declared urgency, written $[U_r - U_d]^+$ . The dangerous direction, because it raises no alert.                                                                                                                                                                                                                                  |
| <b>Milestone truth</b>   | Whether a milestone was met at the deadline originally committed to. Scripted on the first chain; derived from the indicator trajectory on the second.                                                                                                                                                                                                                               |
| <b>Noisy-OR</b>          | Probabilistic aggregation. Reading each block as an independent cause that could on its own put the node in trouble, the aggregate is the probability that <i>at least one</i> of them fires. That is the logical OR of the six, hence the name. Any block approaching its maximum drives the aggregate toward the maximum, unlike a weighted average. See <i>non-compensatory</i> . |
| <b>Non-compensatory</b>  | Property of an aggregation rule in which a healthy criterion cannot offset a failed one. A weighted average is compensatory; the noisy-OR of Equation 1 is not, which is why it was chosen.                                                                                                                                                                                          |
| <b>OEE</b>               | Overall Equipment Effectiveness. Composite of availability, performance and quality, used as the productive health proxy of a resource.                                                                                                                                                                                                                                              |
| <b>Physical Internet</b> | Logistics paradigm in which shipments move through shared, interoperable networks rather than dedicated ones.                                                                                                                                                                                                                                                                        |

|                                 |                                                                                                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PR-AUC</b>                   | Area under the precision-recall curve. More informative than AUC when positives are rare.                                                                                                                                                        |
| <b>Precision@<math>k</math></b> | Among the $k$ nodes a ranking selects, the share that actually fail. The form the question takes when a coordinator can only look at three suppliers.                                                                                            |
| <b>Real urgency</b>             | The urgency computed at a node from six indicator blocks, before any human states anything. Written $U_r$ .                                                                                                                                      |
| <b>Rank / tier</b>              | A supplier's depth in the network, counted from the final customer.                                                                                                                                                                              |
| <b>Ripple effect</b>            | The amplification of a local supply disruption as it propagates through downstream tiers.                                                                                                                                                        |
| <b>SupplyScore</b>              | The tool delivered by this mission. A standalone Python application that computes real urgency, elicits declared urgency, propagates both across a supplier network and scores the gap, running a project week by week with every write audited. |
| <b>Turn</b>                     | One week of simulated time. A campaign turn closes when every node has submitted its weekly review.                                                                                                                                              |

## Résumé

Dans une chaîne d'approvisionnement, l'urgence qu'un opérateur déclare diverge souvent de celle que l'on calcule depuis l'état opérationnel réel. Sur-déclarer gaspille une capacité partagée ; sous-déclarer laisse un risque de rupture invisible. Aucun jumeau numérique, publié ou breveté, ne mesure encore cet écart.

Cette mission, menée à la Direction de l'Innovation d'ALTEN, a conçu et éprouvé une couche logicielle qui le mesure, livrée sous le nom de **SupplyScore**. L'urgence réelle est agrégée depuis six blocs d'indicateurs par une règle *non compensatoire* : le bloc le plus dégradé tire le score vers son maximum, sans qu'un bloc sain puisse l'en empêcher. Elle est ensuite propagée sur le réseau. L'urgence déclarée, elle, est élicitée par un questionnaire de comparaisons par paires contrôlé en cohérence, et l'écart entre les deux est noté sur  $[0, 100]$ .

Dans les tests de *serious game* réalisés, sur les six tours précédant la mise au jour d'une crise, l'écart désigne trois nœuds, tous trois futurs défaillants, sans une seule fausse alerte ; le plus mal noté est signalé huit tours avant de rater son jalon, alors que son responsable déclare une urgence de 0,001.

Trois épreuves indépendantes en ont délimité la portée : elles ont localisé un défaut précis de la couche de prévision, établi que l'outil repère celui qui se méprend sur son état et non celui qui le tait, et mesuré où la simulation apporte réellement sa valeur.

**Mots-clés** : jumeau numérique de chaîne d'approvisionnement, adéquation d'urgence, biais cognitif, aide multicritère à la décision, calibration de prévisions, biais d'automatisation, *serious game*.

## Abstract

In a supply chain, the urgency an operator declares often diverges from the urgency computable from the real operational state. Over-declaring wastes shared capacity; under-declaring leaves rupture risk invisible. No digital twin, published or patented, measures that gap yet.

This mission, carried out at the Direction de l'Innovation of ALTEN, designed and tested a software layer that measures it, delivered as **SupplyScore**. Real urgency is aggregated from six indicator blocks by a *non-compensatory* rule: the most degraded block drives the score toward its maximum, and no healthy block can offset it. It is then propagated across the network. Declared urgency is elicited separately, through a consistency-gated pairwise questionnaire, and the gap between the two is scored on  $[0, 100]$ .

In the serious-game tests carried out, over the six turns before a crisis becomes public, the gap flags three nodes, all three of which later miss a delivery commitment, with no false alarm on any turn; the worst-scoring of them is flagged eight turns before it misses, while its manager declares an urgency of 0.001.

Three independent tests then established its domain: they located a precise defect in the forecast layer, established that the tool detects a declarant who misjudges their own state rather than one who knows it and stays silent, and measured where the simulation genuinely adds value.

**Keywords**: supply chain digital twin, urgency adequacy, cognitive bias, multi-criteria decision analysis, forecast calibration, automation bias, serious game.

## Resumen

En una cadena de suministro, la urgencia que declara un operador difiere a menudo de la urgencia calculable a partir del estado operativo real. Declarar de más desperdicia capacidad compartida; declarar de menos deja invisible el riesgo de ruptura. Ningún gemelo digital, publicado o patentado, mide todavía esa diferencia.

Esta misión, realizada en la Dirección de Innovación de ALTEN, diseñó y probó una capa de software que la mide, entregada con el nombre de **SupplyScore**. La urgencia real se agrega a partir de seis bloques de indicadores mediante una regla *no compensatoria*: el bloque más degradado lleva la puntuación hacia su máximo y ningún bloque sano puede compensarlo. Después se propaga por la red. La urgencia declarada se obtiene por separado, mediante un cuestionario de comparaciones por pares con control de coherencia, y la diferencia entre ambas se puntúa en  $[0, 100]$ .

En las pruebas de *serious game* realizadas, en los seis turnos previos a que una crisis se haga pública, la diferencia señala tres nodos, los tres futuros incumplidores, sin una sola falsa alarma; el peor puntuado se señala ocho turnos antes de incumplir su hito, mientras su responsable declara una urgencia de 0,001.

Tres pruebas independientes delimitaron su alcance: localizaron un defecto preciso en la capa de predicción, establecieron que la herramienta detecta a quien se equivoca sobre su propio estado y no a quien lo conoce y calla, y midieron dónde aporta valor realmente la simulación.

**Palabras clave:** gemelo digital de cadena de suministro, adecuación de urgencia, sesgo cognitivo, análisis multicriterio de decisión, calibración de predicciones, sesgo de automatización, *serious game*.

## Preamble

**Status of the results.** This report is submitted on 24 August 2026, with two months of a six-month internship still to run. One strand is deliberately left open: the tool is being run against additional supply chains to widen the evidence base behind every rate quoted here, and Appendix B states what the remaining time carries. The construction, the measurements and the diagnoses are stable and reproducible from the artefacts listed in Section 4.3. The rates attached to them are those of the campaigns completed to date and may move as that base widens.

**Confidentiality.** The project is designated confidential by ALTEN in the annex to the internship agreement. This report carries a confidentiality marking on every page. It contains internal scoring parameters, the architecture of an unreleased software product and results not published elsewhere.

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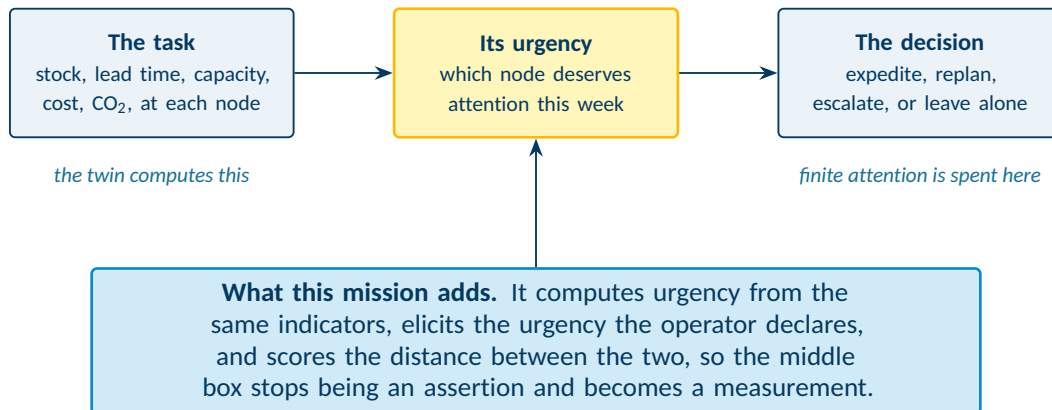
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# 1 Introduction

A supply chain coordinator opens the week with a list of nodes and a finite amount of attention. Somewhere on that list is the supplier that will cause next month's line stoppage. Nothing on the screen says which one it is.

The job has changed shape and the tooling has not followed. A digital twin manages *tasks*: it records what each node has to do, how far along it is, and what state its resources are in. What a coordinator actually has to manage is the *urgency* of those tasks, meaning the order in which a finite amount of attention is spent on them. Figure 1 places the two layers side by side, and this report is about the second one.



**Figure 1:** Where the mission sits. A digital twin measures the state of each task. The decision that consumes a coordinator's week is made on the *urgency* of those tasks, which today no module computes and no module audits.

The reason that middle box stays empty is that the decision inside it is multi-criteria: cost against deadline against supplier reliability against carbon. Supply chains are monitored through indicators that each measure one criterion at a time, so when the tooling cannot combine them, the coordinator combines them by judgment. Judgment is then the real decision layer of the system, and it is the layer nobody measures yet.

The consequences are not evenly distributed. Declaring more urgency than a situation warrants consumes shared transport and production capacity that another flow needed. Declaring less leaves a risk that nobody is watching, and by the time it becomes visible the window to act on it has closed. The second error is the more expensive, and it is the harder one to detect, precisely because an under-declared risk generates no direct signal.

This mission was carried out at the Direction de l'Innovation of ALTEN, within the Smart Green Supply Chain Digital Twin research programme, between 13 April and 16 October 2026. Its subject is the layer that sits before any planning decision: the urgency signal itself. One question drove the work.

*Can the gap between what an operator declares and what the operational state actually supports be measured well enough, inside a working digital twin, to be acted upon?*

Answering it took two steps, and the report is built on that division.

The first step was to build the tool. SupplyScore computes urgency from six indicator blocks, elicits what the operator declares, propagates both across a supplier network, and runs the whole cycle week by week while auditing every write. That step is engineering, and Sections 7.1 and 7.2 report it.

The second step was to find out whether the number the tool produces is worth anything, and that is where the mission's contribution lies. A number a system computes is not yet a number anyone

should trust, so four tests were designed to try to break the tool rather than to show it working. A pre-registered serious-game campaign replayed a documented industrial crisis with independent declarants, and scored the signal against outcomes that were recorded rather than asserted. A second supply chain was then built on deliberately different assumptions, to find out whether the first result was a property of the tool or of the scenario it was born in. A one-line baseline was written to ask whether the most expensive component of the system earns its cost. And a parallel workstream went looking for an indicator no operator supplies at all, estimated from satellite imagery.

All four returned an answer, and in each case that answer proved more useful than the confirmation it might have given instead. What they established, and the boundary each one drew around the tool, is what this report is for.

One result was not planned at all, and it is the one with the widest consequences for anyone deploying such a system. Showing an operator the tool's own forecast, immediately before asking them to declare, changes what they declare. Section 7.4 reports what it changes and in which direction, at two different scales, and Section 8.3 says what an advisory layer owes the person reading it.

The report is organised as the mission was run. Sections 2 to 5 set the company, the problem, the assignment and how the project was run. Section 6 reviews the existing methods, published and patented, and identifies what none of them provides. Section 7 reports the work and what the four tests returned. Section 8 assesses the results against the objectives and closes with a personal assessment against the ESTIA competency framework.

## 2 The Company

### 2.1 The ALTEN Group

ALTEN is a French engineering and technology consulting group founded in 1988, with its head office in Boulogne-Billancourt. It sells engineering expertise rather than products: its consultants are placed on the technical projects of client companies, across the aerospace, defence, energy, transport, telecommunications, health and finance sectors. The group operates in more than thirty countries, and France remains its largest market.

Two features of that model matter for understanding this mission. First, ALTEN's asset is the technical capability of its people, so anything that raises the level of a whole practice has leverage that a single client project does not. Second, a consulting group works for many clients at once, so it sees the same difficulty appear in company after company. That is how a problem gets recognised as belonging to the industry rather than to one customer, and the urgency assessment problem treated in this report is one of those.

Figure 2 places the mission inside the group, from the whole down to the one layer this report is about. The rest of this section walks the same path.



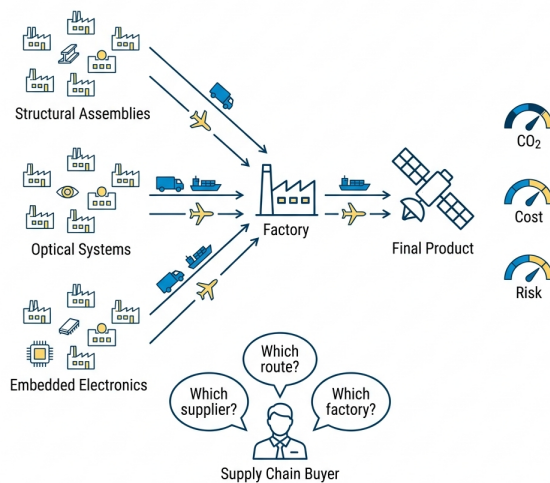
Figure 2: From the group down to the layer this report is about.

## 2.2 The Direction de l'Innovation

The Direction de l'Innovation (DIN) is the entity responsible for ALTEN's own research and development, as distinct from the work its consultants deliver to clients. It operates a network of ALTEN Labs and brings together permanent researchers, doctoral candidates, consultants, interns and apprentices. Its output is not billable hours but reusable assets: methods, prototypes and expertise that consultants and clients can subsequently draw on.

This mission took place at the Toulouse Lab, within the Smart Green Supply Chain research programme, which pursues three aims at once. The lean aim targets waste, meaning cost, delay and excess stock. The green aim targets the carbon footprint of each transport leg, working to the Global Logistics Emissions Council (GLEC) Framework v3.2 (Smart Freight Centre, 2023) and ISO 14083:2023 (International Organization for Standardization, 2023). The predictive aim, where this work sits, targets the anticipation of disruption.

The programme's central artefact is DISCO, a serious game and digital twin simulating a complete supply chain for a satellite product across 22 component suppliers (Baxas, 2025). Figure 3 shows the network players work with and the three dimensions their decisions are scored on.



**Figure 3:** The DISCO serious game network. Players hold one node each and are scored on cost, delay and carbon.

Successive internships have built its components: the Physical Internet routing module (Baulain et al., 2024), the supplier risk scorer (Kabouche, 2025), the per-route carbon quantification (Cordova, 2024) and the transport management interface (Mondelice, 2025). The work described here adds a layer none of them covers, and it ran alongside a parallel internship on the network representation and geographic modelling side of the same programme, discussed in Section 5.1. DISCO is the environment the present work was designed to be compatible with, although the delivered system runs alongside it rather than inside it, for reasons given in Section 4.2.

## 2.3 Politique RSE d'ALTEN (CSR policy, in French)

ALTEN a adhéré au Pacte Mondial des Nations Unies en 2010 et structure depuis sa politique de responsabilité sociétale autour de trois piliers : l'humain, l'environnement et l'innovation durable. La démarche se mesure autant qu'elle se déclare : score EcoVadis de 85/100 en 2025, niveau Platinum et premier centile mondial, note A- confirmée au Carbon Disclosure Project, et

trajectoire de réduction validée par la Science Based Targets initiative en 2023 avec une neutralité carbone visée à l'horizon 2050.

Sur le pilier humain, le Groupe réunit plus de cent nationalités, a signé les *Women's Empowerment Principles* des Nations Unies et publie au titre de 2025 un index d'égalité professionnelle de 88/100 pour ALTEN Sud-Ouest, l'entité juridique dans laquelle s'est déroulée cette mission ; le programme INSPIRE 2026-2030 en porte la suite. Sur le pilier environnemental, les actions internes sont concrètes et vérifiables : 100 % d'électricité renouvelable pour les bâtiments français dont la puissance souscrite dépasse 36 kVA, 80 % des surfaces couvertes par le tri sélectif, une certification ISO 14001 en cours d'extension, et des ateliers Fresque du Climat.

C'est le troisième pilier qui éclaire le mieux cette mission. ALTEN consacre 31 % de sa R&D à l'innovation environnementale, et le programme dans lequel ce stage s'est inscrit en fait partie : un programme *Smart Green Supply Chain*, où les émissions de CO<sub>2</sub> constituent l'un des six blocs du modèle d'urgence développé ici, au même rang que le coût ou le délai, et non une correction appliquée après coup. La politique RSE du Groupe n'a donc pas seulement encadré ce travail : elle en a déterminé une variable de modélisation.

**Prise de recul.** Le carbone est l'un des six blocs du modèle, et c'est pourtant celui qui a le moins servi pendant la première campagne. La raison n'est pas un choix de modélisation, c'est un manque de données. Les émissions par trajet sont connues à l'échelle du programme, pas encore à celle d'un nœud et d'une semaine, qui est la maille où l'urgence se calcule. Une ambition environnementale peut donc être inscrite dans un modèle et y rester la dimension la moins active. Avoir montré à quel endroit précis la donnée manque est, à mon sens, la contribution RSE la plus utile de cette mission.

### 3 Problem Statement

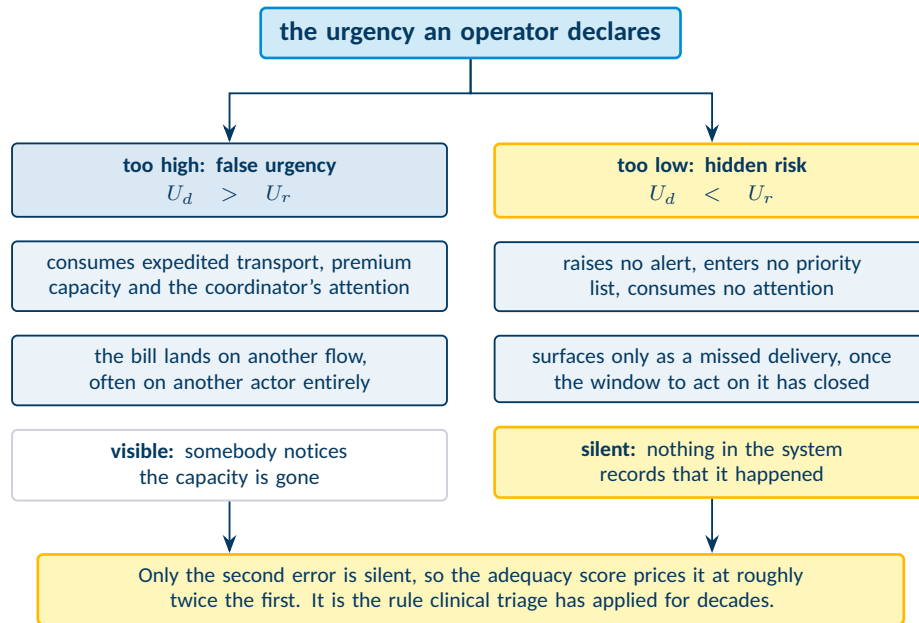
#### 3.1 Why ALTEN Opened This Subject

Every component the programme had built by the start of this internship answers a question of the form *what is the state of this part of the chain?* None answers the question that comes next. Once the state is known, a coordinator still has to decide where to spend a finite amount of attention this week, and that decision runs on a signal none of those modules produces: how urgent each node is. In practice it comes from people. A buyer says a supplier is critical; a plant says a line is at risk. The twin records the consequence of those judgments without ever evaluating the judgments themselves.

That is a gap with a cost attached, because the signal is not neutral. Zhu, Yang and Hsee demonstrate across five controlled experiments that people systematically favour imminent low-value tasks over distant high-value ones, and that the bias persists even when they are told about it (Zhu et al., 2018); field studies find the same pattern under workload (Rusou et al., 2020; DiwasSingh et al., 2020). A twin that consumes a human urgency declaration as ground truth inherits that bias and propagates it across the network.

#### 3.2 The Cost of the Two Errors

The two directions of error are not symmetric, which is what makes the problem worth instrumenting rather than tolerating. Figure 4 sets them side by side.



**Figure 4:** The two directions in which a declaration can be wrong.

Both costs are documented. In a shared or mutualised logistics setting the price of over-declaring lands on another actor entirely (Montreuil, 2011; Li et al., 2020), and clinical triage, where the asymmetry has been studied for decades, reaches the same verdict on the other direction: under-triage costs a life where over-triage costs a queue (Kim et al., 2020; Mackway-Jones et al., 2013). That asymmetry is the argument behind the penalty of Section 7.1.

### 3.3 The Problem, Stated

#### Problem statement.

A supply chain digital twin can compute the physical state of every node, and it can record the urgency a human declares for each of them. It does not compare the two. The declared signal is known to be biased, the computed signal is available, and the difference between them is exactly the quantity that would tell a coordinator which node is being mismanaged rather than merely which node is struggling.

*Can the gap between declared and computed urgency be measured reliably enough, inside a working digital twin, to be acted upon rather than merely observed?*

The word *reliably* carries the weight. Producing a number that expresses the difference between two other numbers is trivial. Establishing that the number means something, and that it predicts outcomes better than the raw indicators a coordinator already has, is the actual engineering research problem.

## 4 The Mission

### 4.1 The Assignment as Written

The internship agreement defines the subject as *Stage Innovation: Ingénieur Supply Chain* and the assignment as the update and development of a supply chain digital twin, over the five workstreams Figure 5 names. It further specifies an innovation project with a state of the art, a scientific report,

multidisciplinary project management and an evolutionary approach based on Agile and Test and Learn.

That scope is deliberately broad, which is normal for an R&D assignment and is itself a managed risk (Section 5.5). The first weeks were therefore spent converging it onto a defensible contribution, in agreement with the research programme director. Converging is not narrowing: Table 1 states what the mission delivered against each of the five.

**Table 1:** The five signed workstreams and what the mission delivered against each

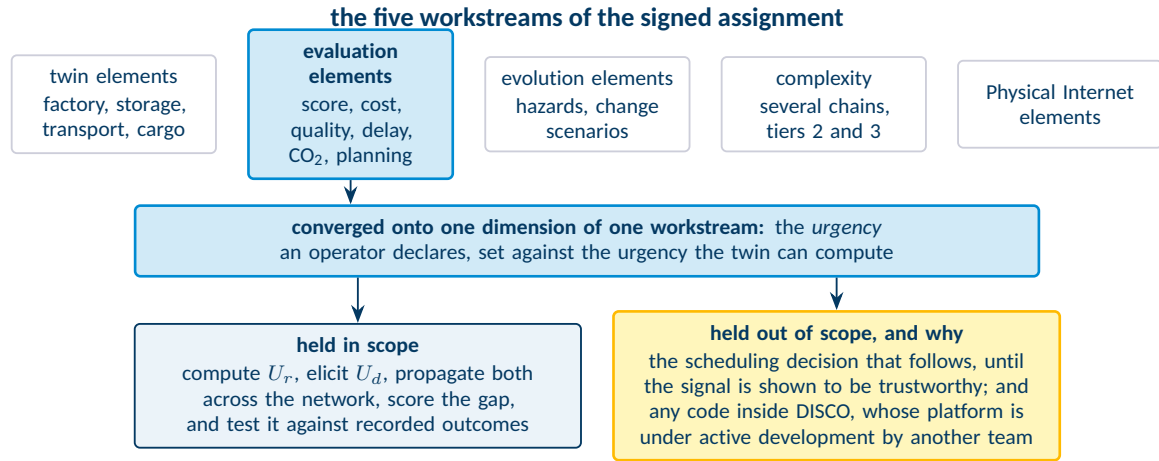
| Workstream                                                                   | Delivered                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Twin elements: factory, storage, transport, cargo                            | Storage capacity was attacked directly, and by observation rather than declaration: a segmentation model estimates a warehouse’s physical capacity from satellite imagery (Section 7.8). Factory and transport nodes carry the indicator set the model consumes |
| Evaluation elements: score, cost, quality, delay, CO <sub>2</sub> , planning | The converged core. All six became blocks of the urgency model, delay as time and quality as performance, alongside capacity, risk, cost and carbon, aggregated into one score and compared against the declared value (Section 7.1)                            |
| Evolution elements: positive and negative hazards, change scenarios          | An event engine with twelve calibrated events, a scenario variant generator, and two complete campaigns that replay them turn by turn (Sections 7.3 and 7.6)                                                                                                    |
| Complexity: several supply chains, tier-2 and tier-3                         | Two chains built and run, of three and five tiers, 23 nodes in total, with propagation across every tier (Section 7.6)                                                                                                                                          |
| Physical Internet elements                                                   | Held by the programme’s existing routing module (Baulain et al., 2024). The urgency layer is designed to sit above it and deliberately does not modify it                                                                                                       |

4.2 The Converged Objective

The evaluation workstream is where the twin was weakest, because it computed state without ever assessing the human judgment applied to that state. The mission converged on that workstream, and specifically on the urgency dimension of it.

**Converged objective.** Design, implement and empirically evaluate a decision-support layer that computes a real urgency from the twin’s operational state, captures the urgency declared by the operator through a structured elicitation, propagates both across the supplier network, and scores the gap between them in a way that can be tested against recorded outcomes.

Two boundaries were then set explicitly and held. Figure 5 records them alongside the convergence itself.



**Figure 5:** Converging did not drop the other four workstreams (Table 1 states what each received). It fixed which one the mission would be judged on.

One workstream from the signed scope was kept alongside the converged objective rather than dropped. The twin’s capacity element depends today on a figure the operator types in, which makes it the one indicator the adequacy measurement cannot treat as independent. A parallel effort therefore estimates warehouse capacity from public satellite imagery (Section 7.8). It is reported as an opening rather than as a delivered capability, and it carries its own objective in Table 2.

### 4.3 Objectives, Indicators and Deliverables

Table 2 states the objectives agreed with the programme director, each with the indicator used to judge it and the deliverable that carries it. The indicators were chosen so that each could be answered yes or no from an artefact rather than from an opinion.

**Table 2:** Objectives, indicators and deliverables

|    | Objective                                                                                                                    | Indicator                                                                                                       | Deliverable                                                   |
|----|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| O1 | Define a real urgency computable from heterogeneous operational indicators, and an adequacy score against the declared value | A model whose every term is decomposable, and whose aggregation choice is justified against the alternatives    | Mathematical specification                                    |
| O2 | Deliver a working, persistent, auditable application implementing it                                                         | Runs multi-tenant; every write audited; automated test suite passing                                            | SupplyScore application                                       |
| O3 | Propagate urgency across a multi-tier network and rank nodes by criticality                                                  | Handles a network at the scale of a real programme within an interactive time budget                            | Propagation and criticality services                          |
| O4 | Establish whether the resulting signal predicts recorded outcomes                                                            | A score against a stated outcome definition, compared with a trivial baseline                                   | Pre-registered experimental protocol and its results          |
| O5 | Capitalise the research so the programme can continue it                                                                     | A written scientific record and reproducible derivation of every number                                         | <i>Bilan Scientifique et Technique</i> and derivation scripts |
| O6 | Establish whether warehouse capacity can be observed rather than declared                                                    | A physical feasibility result, and a model scored on the operational quantity rather than on a detection metric | Segmentation model, annotated dataset and its audit           |

**Intermediate deliverables.** Four were produced during the mission rather than at its end: a knowledge transfer report on planning and urgency, delivered in week 2 and used to align vocabulary with the programme; a state of the art with an accompanying presentation, delivered in week 3; the mathematical specification, reviewed before implementation began; and the frozen experimental protocol, hashed and registered before any campaign data existed, so that the hypotheses could not be adjusted after seeing results.

**Final deliverables.** The SupplyScore application with its test suite and documentation; the experimental campaign with its two executed arms and their analysis; the *Bilan Scientifique et Technique*; and the derivation scripts that regenerate every reported figure from raw artefacts.

**The fourth objective became the centre of the mission.** O4's indicator required the signal to be compared against a trivial baseline rather than judged in absolute terms. Holding to that clause turned a validation step into the main scientific contribution of the internship, and produced the three tests of Section 7.

## 5 Project Organisation

### 5.1 Stakeholders and Responsibilities

The mission was carried out inside a small research team, and the distribution of responsibility mattered more than its size suggests. Table 3 lists who did what.

**Table 3:** Stakeholders and their role in the mission

| Person                           | Role                                                       | Responsibility in this mission                                                                                                                           |
|----------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sébastien Perthuisot             | Research Programme Director, Quality & Supply Chain Expert | Set the research direction, arbitrated scope, chaired the weekly regular point and the Friday demonstration. Decision-maker on what the programme keeps. |
| Émilienne Lardy and Selma Ferhat | Scientific supervision                                     | Ran the Monday planning and the three weekly scrums; reviewed method, scientific framing and written output. First readers of anything claimed.          |
| Liam Jones                       | Fellow intern, same programme                              | Network representation, graph modelling and geographic layer of the twin. Neighbouring problem, shared demonstration slot.                               |
| David Gomez                      | ESTIA academic supervisor                                  | Academic framing of the mission.                                                                                                                         |
| John Hoarau (author)             | Intern, Supply Chain Engineer                              | Model, implementation, experimental design, analysis and written deliverables.                                                                           |

The presence of a second intern on an adjacent problem shaped the work more than expected. Liam Jones was building the structural and geographic representation of a supply chain, meaning how a network is described, generated and visualised, while this mission built the layer that scores nodes once they exist. The two are complementary halves of the same twin, and the Friday demonstration forced each to state weekly what the other could rely on. It also produced a boundary decision early: the scoring layer would assume a network is given rather than attempt to generate one, because that was being solved two seats away.

### 5.2 Resources

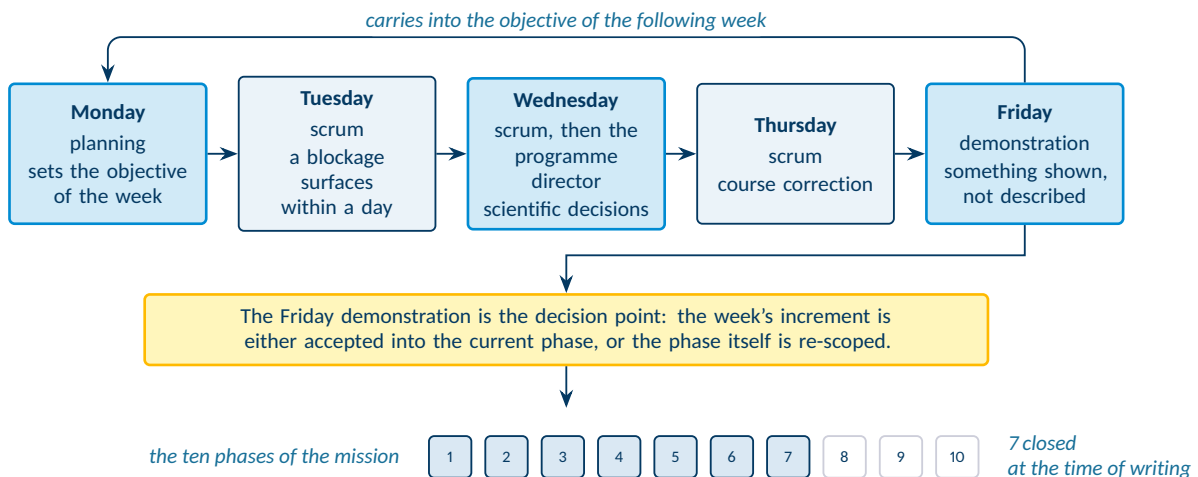
The resources were modest, and one of them constrained technical choices. All implementation was carried out by the author, on one ALTEN laptop, a Ryzen 7 PRO 8840HS with 16 cores and no discrete

GPU, on which everything in this mission was computed, every model training run included. Azure DevOps carried source control and work tracking. No budget was managed, and none was requested: no paid data service, no cloud compute, no commercial solver. That absence removed managed services and paid datasets from the solution space before any technical comparison took place, which is why the delivered system depends only on open-source Python libraries and a file-based database.

### 5.3 Working Method: Agile and Test and Learn

The internship agreement prescribes an evolutionary approach based on Agile and Test and Learn. That was applied literally rather than nominally, and the weekly rhythm was the mechanism.

**The cadence.** The week had a fixed shape, and Figure 6 gives it together with the way it nests inside the mission plan. Four contact points a week on a six-month mission is a heavy cadence for R&D, and it was the single most useful organisational feature of the internship: it made it impossible to spend three weeks on an idea that would not survive contact with a reviewer. It is also the mechanism that ties the week to the plan. A phase advances by one week's increment, and the Friday demonstration is where that increment is accepted or the phase is re-scoped, which is how the reallocation described in Section 5.4 was decided.

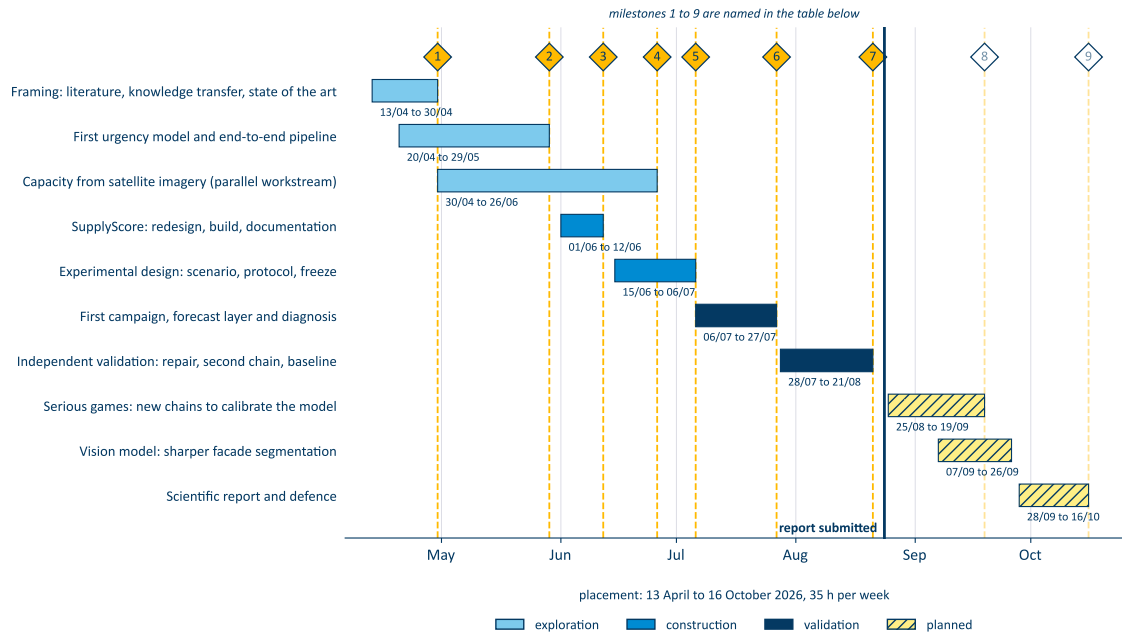


**Figure 6:** The weekly cycle, and how it nests inside the mission plan. Shaded days carry the meetings that can change the plan: Monday planning, the Wednesday point with the programme director, and the Friday demonstration. Each of the ten phases advances by whole weeks of this cycle and closes on a dated milestone artefact (Table 4); the three pale phases at the end are the ones still to run at the time of writing.

**Test and Learn, and what it rejected.** The method means something only if things were abandoned. Four were. The first urgency formulation went in April, for the reason given in Section 7.7. The May pipeline of separate scores was superseded in June by the design that shipped, having first established what the interfaces between components needed to be. A third-party logistics dataset considered for the freight node was rejected on inspection, its pre-computed target columns, misplaced coordinates and uniformly flat distributions all signatures of synthetic data; the node was driven by calibrated events instead. And the human arm of the campaign was reduced to isolated agents when eight consultants for nine days could not be secured, which weakened every behavioural claim rather than the design (Section 5.5).

### 5.4 Phases and Milestones

The mission runs in ten phases, seven of them closed at the time of writing. Figure 7 gives the schedule and Table 4 the milestones that closed each phase.



**Figure 7: Phases and milestones of the mission**

**Table 4: Milestones**

|   | Date      | Milestone                     | Closed by                                                                                                                                          |
|---|-----------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 30 April  | State of the art delivered    | Knowledge transfer report on planning and urgency (20 April), state of the art and its presentation; subject converged with the programme director |
| 2 | 29 May    | First pipeline running        | Six modules and the unified score computing end to end, found insufficient, which triggered the redesign                                           |
| 3 | 12 June   | SupplyScore v2.0.0            | Seventeen delivery phases: multi-tenant persistence, audited writes, weekly cycle, event engine, explainability, exports, documentation            |
| 4 | 26 June   | Capacity from imagery         | Dock counting shown unreachable from nadir imagery, problem reframed to facade segmentation, four training runs scored                             |
| 5 | 6 July    | Protocol frozen               | Scenario, hashed data pack, role cards and hypothesis register, all before any campaign data existed                                               |
| 6 | 27 July   | First campaign scored         | Two arms analysed against the pre-registered register; the forecast layer's defect located                                                         |
| 7 | 21 August | Independent validation closed | Repair re-measured on frozen forecasts; second chain run in both arms; baseline comparison scored                                                  |
| 8 | 19 Sept   | Calibration campaigns         | <i>Planned.</i> New chains to widen the evidence base                                                                                              |
| 9 | 16 Oct    | Scientific report             | <i>Planned.</i> Bilan Scientifique et Technique and defence                                                                                        |

Seven phases are closed, two of them beyond the campaign that was originally the mission's end point. The three that remain carry no new construction. Phase boundaries are read from the dated milestone artefacts rather than from commit timestamps, since work was pushed in batches: seventeen commits landing on 11 June close a fortnight of work.

## 5.5 Risks and Mitigations

Four risks were carried from the start, and Table 5 states each one, whether it materialised and what was done about it. Three of the four did materialise.

**Table 5: Risk register and outcomes**

| <b>Risk</b>                                           | <b>Outcome</b> | <b>Mitigation and effect</b>                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Literature access routed through university libraries | Materialised   | Retrieving a paper on ESTIA and Cranfield credentials was never a same-hour operation. Mitigated by front-loading retrieval into the first three weeks against a planned list, so no design decision waited on a download.                                                                                                                                                                                             |
| Broad mission scope                                   | Materialised   | Five workstreams against one person and six months. Mitigated by converging the subject with the programme director and recording the boundaries explicitly (Section 4.2).                                                                                                                                                                                                                                             |
| Human campaign might not be securable                 | Materialised   | Eight colleagues for nine working days was never guaranteed. Mitigated by a protocol in which the declarant could be substituted without changing the record format: the substitute is a language-model agent answering the same questionnaire behind the same consistency gate, so an incoherent weight vector is refused exactly as it would be for a human. Cost: the external validity of every behavioural claim. |
| CPU-only compute                                      | Contained      | No GPU available, which bounded model size and training resolution and is why one comparison ran at reduced scale. Mitigated by choosing methods whose cost is dominated by data rather than by parameters.                                                                                                                                                                                                            |

## 6 State of the Art

The question of Section 3.3 sits at the meeting point of several fields that rarely cite one another. This section reviews each for what it settles and, more usefully, for what it leaves open. A structured appraisal of every source, recording its argument, its method and its limits, was maintained throughout the mission and informs the readings below.

### 6.1 Deadline-Driven Value and Risk Propagation

Real-time systems research models the value of a task as a function of when it completes. Chen and Mühlethaler show that utility typically holds constant until a deadline and then decreases (K. Chen et al., 1996), and a wider survey confirms structured degradation past critical thresholds (J. Chen et al., 2025). This establishes that urgency is legitimately time-varying.

Separately, supply chain research formalises the ripple effect, whereby a local disruption cascades through downstream tiers and amplifies as it travels (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020). This establishes that urgency is legitimately network-dependent.

*Appraisal.* The first stops at a single task, the second takes the node-level score as given, and neither addresses a declared signal. Together they justify treating urgency as time-varying and propagated, and leave aggregation open.

### 6.2 Indicators of Delivery Rupture

If urgency is to be aggregated, the indicators must come from somewhere. Heckmann, Comes and Nickel establish rupture as the terminal event a supply risk measure should anticipate, and criticise the field for measuring exposure without connecting it to that event (Heckmann et al., 2015). Indicator-based supplier monitoring proceeds by identifying observable precursors (Mokhtar et al., 2019).

Five families recur. Slack is the only indicator whose exhaustion is irreversible (Cohen et al., 2019; Sun et al., 2020). Capacity saturation adds waiting time a nominal lead time does not show (Tomlin,

2006; Hosseini; Tajik, et al., 2019). Resource performance is monitored through composite proxies because supplier-side data is rarely available (Mokhtar et al., 2019; Hlioui et al., 2017). Adverse-event exposure is treated probabilistically (Hosseini; Ivanov, et al., 2020; Liu et al., 2021). Cost signals are read as symptoms of strain rather than causes (Shen et al., 2019), and emissions constraints restrict the admissible responses rather than causing rupture (Hoen et al., 2014; Smart Freight Centre, 2023).

*Appraisal.* This literature monitors precursors one at a time. It gives no criterion for which belong in one composite, nor whether equal values on structurally different indicators carry equal risk meaning; Section 7.1 treats the second.

### 6.3 Cognitive Bias in Declared Urgency

Zhu, Yang and Hsee report the mere-urgency effect across five controlled experiments: participants consistently preferred imminent low-value tasks to distant high-value ones, and the bias survived being told about it (Zhu et al., 2018). Steel gives a formal discounting model in which immediacy becomes more salient as a deadline approaches (Steel et al., 2006). Field studies confirm the pattern under workload (Rusou et al., 2020; DiwasSingh et al., 2020).

*Appraisal.* Quantified in controlled settings, uncalibrated in industrial ones. It justifies treating the declared signal as biased without saying what to do about it.

### 6.4 Multi-Criteria Elicitation and Aggregation

Structured elicitation of a human judgment is a solved problem in outline. The Analytic Hierarchy Process derives criteria weights from pairwise comparisons and validates them through a consistency ratio (Saaty, 1980), though it is vulnerable to rank reversal and to cognitive noise (Dyer, 1990; Triantaphyllou et al., 2024). The Best-Worst Method reduces the number of comparisons required (Rezaei, 2015), and its fuzzy extension absorbs judgment uncertainty (S. Guo et al., 2017; Liang et al., 2020). For ranking alternatives without letting a good score on one criterion mask a catastrophic one on another, outranking methods such as PROMETHEE II establish pairwise preference flows (Brans et al., 1986; Behzadian et al., 2010).

The composite indicator literature supplies the necessary warning: nominal weights often diverge from effective importance because of correlation and normalisation effects (Greco et al., 2018; Becker et al., 2017; Borgonovo et al., 2021).

*Appraisal.* These are static tools: they elicit and aggregate, and none is routinely compared against a computed counterpart. The composite-index critique stays a closing caveat rather than something a running system checks on its own data.

### 6.5 Digital Twins and Decision Support

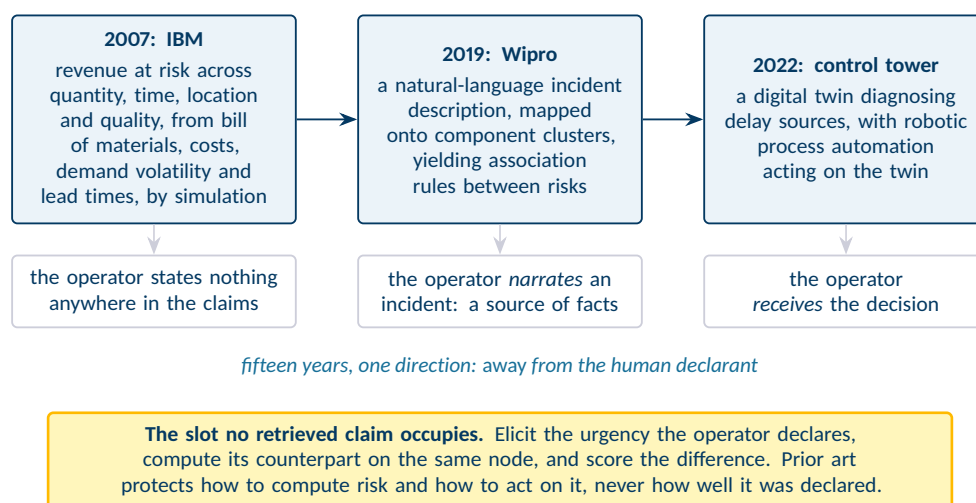
The digital twin concept was formalised by Grieves (Grieves, 2014) and extended to manufacturing (Tao et al., 2018; Fuller et al., 2020). In supply chains it integrates simulation, optimisation and machine learning (Ivanov; Dolgui, et al., 2021), with demonstrated capability in disruption monitoring and replanning (Ding et al., 2023; Leung et al., 2022; Hrušovský et al., 2021). Recent human-in-the-loop cognitive twins capture human feedback to synchronise mental models with simulated operations (Niloofar et al., 2023).

*Appraisal.* This is the family the delivered system belongs to, and the gap is here. No retrieved system combines a declared urgency, a computed one and their propagated mismatch. Where an input depends on the operator the system exists to support, that circularity is not treated as a measurement problem.

## 6.6 Industrial Solutions and Patents

Academic absence would be weak evidence on its own, since a working technique can be commercialised without ever being published. The patent record was therefore searched for the same combination.

Three sustained lines emerged, and Figure 8 sets them out. IBM's quantifies revenue at risk across the quantity, time, location and quality dimensions of a disruption, entirely from measured or forecast inputs (Corporation, 2007). Wipro's replaces those inputs with language, extracting entities from a natural-language incident description and deriving association rules between risks (Limited, 2019); the human appears as a narrator, and nothing in the method compares what they say against a computed counterpart. The most recent is a digital twin control tower diagnosing delay sources and predicting maintenance needs across replicated value chain entities, with robotic process automation acting on the twin (LLC, 2022).



**Figure 8:** The three sustained patent lines, ordered by what each does with the human. As the claims get more autonomous, the declarant is progressively written out, and the quantity this mission measures is never claimed.

*Appraisal.* The absence extends past the patent record. Commercial offerings in the same space document risk scoring and event alerting rather than declaration quality.

## 6.7 Verifying a Probabilistic Signal

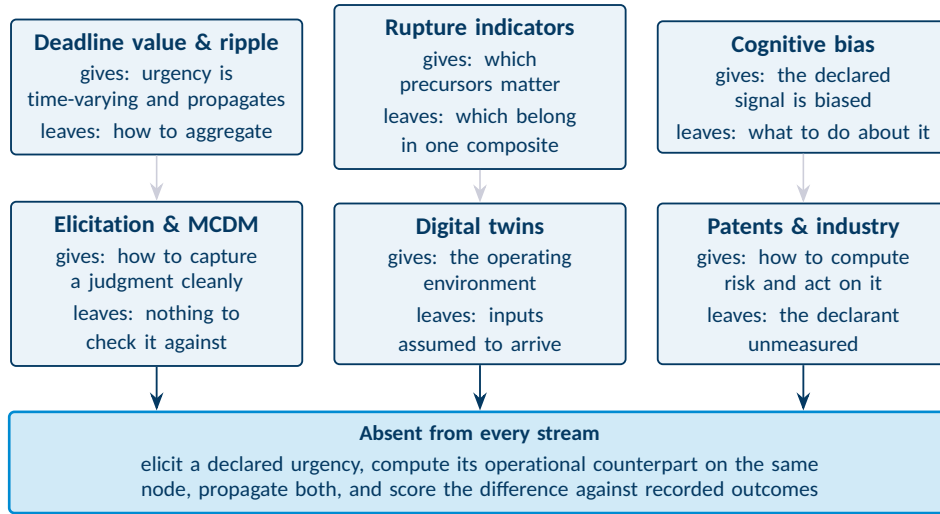
Because this mission scores a probabilistic signal against outcomes, it inherits forecast verification. Brier introduced the mean squared error of probabilistic forecasts (Brier, 1950), and Murphy decomposed it into reliability, resolution and uncertainty (Murphy, 1973), establishing the distinction the results of this report depend on: a forecast can order outcomes correctly while announcing the wrong probabilities, and the two failures need different repairs. Machine learning rediscovered the same point, finding strong classifiers that produce distorted probabilities correctable by a cheap post-hoc layer (Niculescu-Mizil et al., 2005; C. Guo et al., 2017).

A second convention is inherited with it. A verification score is read against a reference forecast, not in absolute terms, and the reference is chosen to be the cheapest thing a practitioner could have done instead: climatology, persistence, or a one-line heuristic (Murphy, 1973). Section 7.6.4 applies that convention to the layer built here, and it is the most demanding test in this report.

*Appraisal.* It assumes the outcome is unambiguously defined. In an operational supply chain the outcome is a modelling choice, and Section 7.4.5 measures that choice moving the reported result by a factor of six.

## 6.8 What the Review Establishes

Each stream closes on a limit, and the limits describe one absence rather than several. Figure 9 places them side by side.



**Figure 9:** Six streams, what each settles and what each leaves open. The lower box is the combination none of them provides, and the one this mission builds.

No retrieved work, published or patented, combines those streams with a structured declared-urgency input and an empirical test of the resulting signal against recorded outcomes. That combination is what this mission built, and then bounded.

## 7 Carrying Out the Mission

This section follows the phases of Section 5.4: the model, the system built from it, a parallel workstream that observes capacity instead of asking for it, and the three tests run against the system. The problems met along the way close the section.

### 7.1 The Model

#### 7.1.1 Real urgency

Real urgency is not read from a single deadline-proximity signal. It is aggregated at each node from six indicator blocks selected from the literature of Section 6: time, capacity, performance, risk, cost and carbon. Each produces a partial urgency  $u_m \in [0, 1]$ .

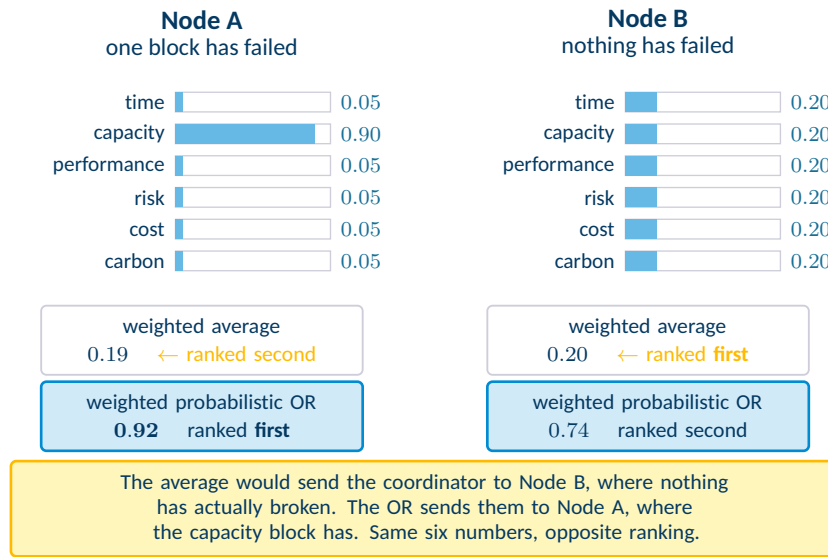
**What the OR means, and why it is one.** Read each block as an independent cause that could on its own put the node in trouble, and read  $u_m$  as the probability that it does. The node is then in trouble if the time block fires *or* the capacity block fires *or* any of the four others does. The probability of that disjunction is one minus the probability that none of them fires, which is exactly the form the model uses, with an exponent  $\omega_m$  setting how much each block is allowed to contribute:

$$U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}. \quad (1)$$

The OR in the name is therefore the logical connective, not an arithmetic operation; the literature calls this form a *noisy-OR*. Choosing it over a weighted average is the central modelling decision of the mission, because the two behave in opposite ways on the situation the tool exists to catch. A weighted average is fully *compensatory*: healthy cost, carbon, and performance readings average away a capacity that has failed outright, and the node scores as moderately urgent. Equation 1 does not compensate,

since a block approaching its maximum drives the aggregate toward the maximum whatever the others say.

Figure 10 works both rules through on two nodes, and they rank them in opposite orders.



**Figure 10:** The same six block values under the two candidate rules, with  $\omega_m = 1$  throughout.

For a signal whose purpose is to catch the node that is about to fail, non-compensatory behaviour is the required property: the coordinator has to be sent to the node where something has broken, not to the node where nothing has.

**Table 6:** The six indicator blocks and what each measures

| Block       | Role         | What it measures                                                                                                                        |
|-------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Time        | Direct cause | Probability that the realistic lead time exceeds the margin to the next deadline. The only resource whose exhaustion cannot be reversed |
| Capacity    | Direct cause | Saturation of storage, weight and throughput, which adds waiting time a nominal lead time does not show                                 |
| Performance | Direct cause | Drift in the productive health of the resource making the part, through a composite availability, performance and quality proxy         |
| Risk        | Direct cause | Expected unavailability: failure probability times recovery time times severity, revised weekly from declared events                    |
| Cost        | Arbitration  | Financial strain. Paying premium freight to protect a deadline is a symptom, not a cause, and it bounds which responses stay affordable |
| Carbon      | Arbitration  | Emissions above target. Restricts the admissible set of responses, typically in the sea-against-air trade-off                           |

Table 6 gives the six. Four describe a mechanism that leads to rupture; two constrain which responses remain available once it threatens. Each candidate was screened on four questions: a plausible causal link to rupture, measurable or credibly estimable, open to operator action, and adding information the retained blocks do not already carry. A block with no data carries weight  $\omega_m = 0$ , kept distinct from  $u_m = 0$ : the first says nothing is known, the second asserts the block is healthy.

**The blocks are treated as exchangeable, and they are not.** The product is symmetric, so it ranks the six blocks only by weight. Yet  $u_{\text{time}} = 0.3$  is a probability of missing a deadline while  $u_{\text{cost}} = 0.3$  is an ordinal position wearing a cardinal number. The consequence was carried rather than hidden:

the aggregate is used for ranking and for comparison against the declared value, never as a calibrated probability.

### 7.1.2 Declared urgency and the adequacy score

Declared urgency is not collected as a free-text estimate. At each node the operator compares four criteria pairwise through AHP, which yields a consistency-checked weight per criterion, then rates each on a severity scale. The four criteria were chosen so that each has a computed counterpart: operational impact and downstream dependencies mirror the severity the propagation amplifies, the time window mirrors the time block, and recoverability mirrors the recovery term of the risk block. Without that correspondence the later comparison would be between unrelated quantities.

Figure 11: The pairwise comparison questionnaire, as the operator meets it.

A weight vector whose consistency ratio exceeds the threshold is refused, and the operator is sent back to the most contradictory of their own comparisons (Figure 11). Without that gate an incoherent judgment would propagate as though it were a considered one.

Both signals then propagate across the network in opposite directions, declared urgency descending from the customer toward suppliers and computed risk ascending the other way, since demand pressure is what a customer transmits and physical risk what a supplier transmits. Their mismatch splits into the two pathologies of Section 3.1, and the gap is scored on  $[0, 100]$  through a penalty that prices hidden risk at roughly twice false urgency. Figure 12 shows both signals over the life of a task.

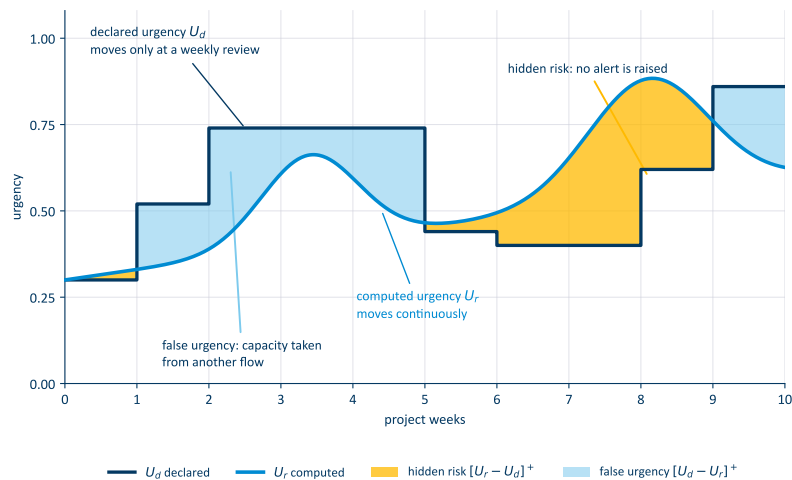


Figure 12: Declared and real urgency over the life of a task.

## 7.2 The System

The model was delivered as SupplyScore, a standalone Python application driving a multi-page web interface over a per-node database layer. It runs a full project week by week: onboarding a network, collecting the weekly reviews, applying events, recomputing every score and exporting the result. Figure 13 gives the software architecture, and the four decisions that carry the engineering weight are visible in it.

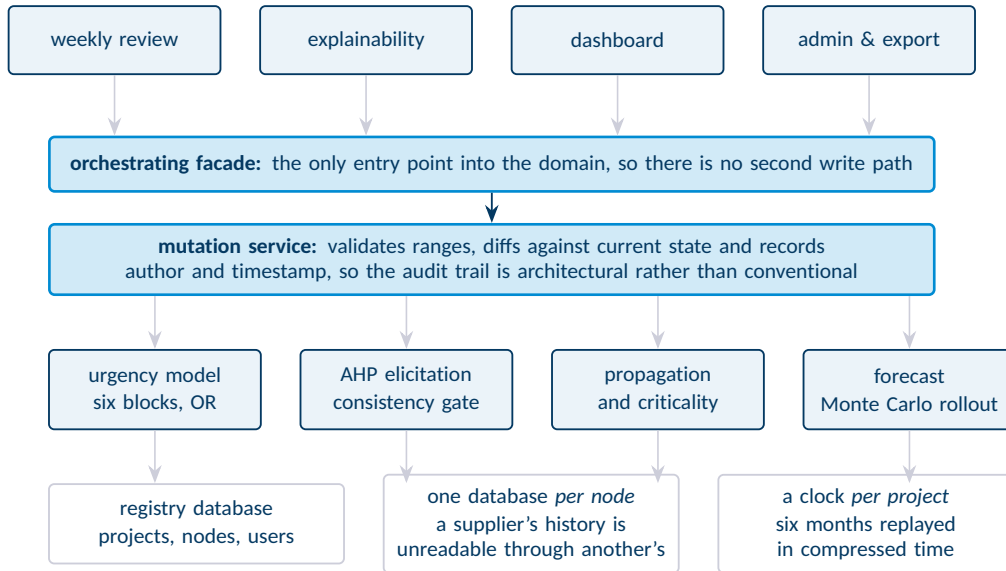


Figure 13: SupplyScore's software architecture.

Figure 14 gives the domain model those services operate on.

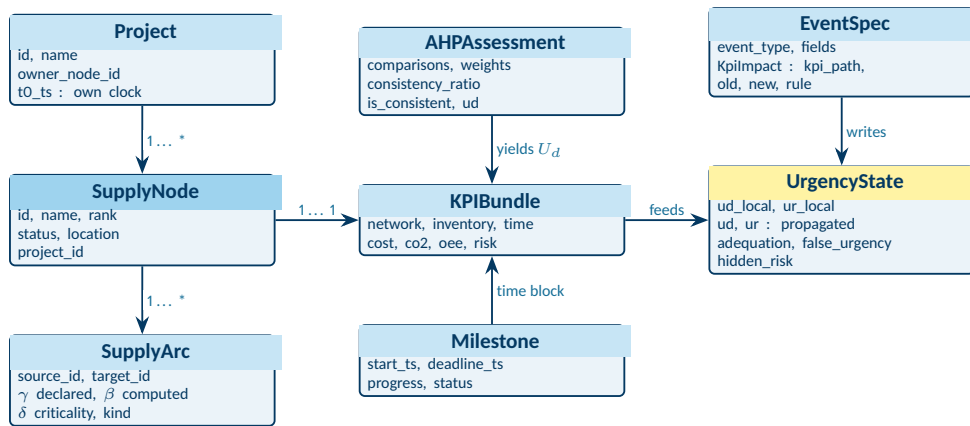


Figure 14: The domain model. A project owns nodes, each node carries one indicator bundle and one urgency state, and the questionnaire, the milestones and the event engine are the three things that write into them.

The operator's entry point is the weekly review page. Its four sections, the questionnaire, the week's indicators, milestone progress and any events, commit as one signed transaction, so no partial review ever enters a node's history.

Two capabilities earned their cost. The explainability page reconstructs any score term by term (Figure 15), which is what made the diagnosis in Section 7.4 possible from the application rather than by hand. The export produces frozen data, which the whole experimental analysis relies on.

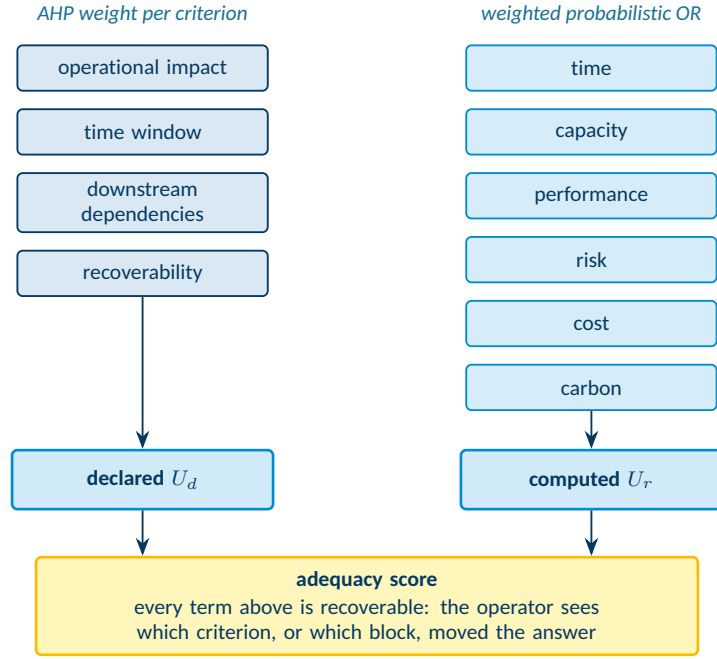


Figure 15: Per-node explainability.

### 7.3 The First Campaign

Building a system that computes a number establishes nothing about whether the number means anything, so the experiment was designed before the system was finished. The campaign replays the 2020 to 2022 semiconductor shortage over nineteen turns across an eight-node satellite supply chain. Real urgency is fed from public time series of the actual crisis; declared urgency comes from the participants through the weekly questionnaire, and is taken at face value as a sincere report of what its author perceives. That hypothesis is what makes the two signals independent in source, which is the precondition for the gap between them to mean anything.

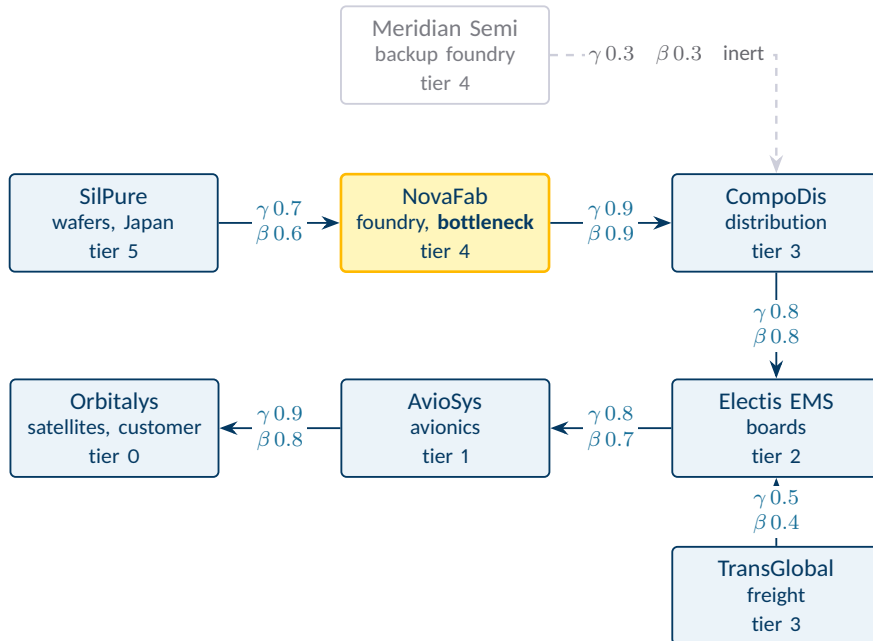


Figure 16: The first campaign network.  $\gamma$  attenuates declared urgency as it descends toward suppliers,  $\beta$  amplifies computed risk as it ascends toward the customer.

The foundry at the deepest tier is the expected bottleneck, and the campaign is built so the tool has to find it without being told (Figure 16).

Four artefacts were frozen before any data existed: the scenario with its twelve calibrated events, the data pack hashed per file, the role cards, and a register of six hypotheses each carrying a numeric threshold. Freezing the thresholds in advance prevents a disappointing result from being rescued by moving the bar afterwards.

The design compares two arms on the identical frozen scenario. In the control arm, eight isolated declarants each see only their own role card and turn briefing and declare blind. In the treatment arm, the same declarants are additionally shown the system's own predictive analysis of their node before declaring, and record whether it influenced them. Because the scenario is frozen, the forecasts are identical between arms, so any behavioural difference belongs to the act of displaying them.

## 7.4 Results on the First Chain

The results divide in two, reported in that order and both re-examined on a second chain in Section 7.6. The system does what it was built to do, and the predictive layer on top of it was scored, its limiting component located, repaired and re-measured.

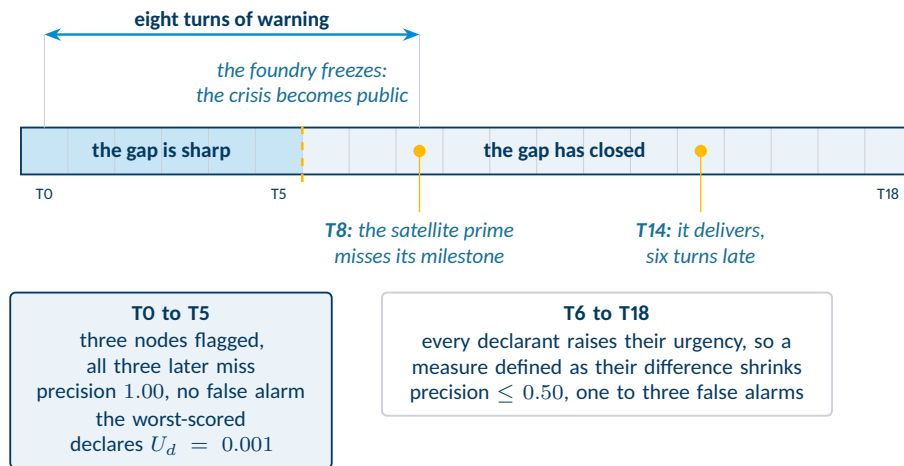
### 7.4.1 The adequacy signal identifies the right nodes, and does it early

A coordinator does not ask whether node  $X$  will suffer an adverse event between weeks  $t + 1$  and  $t + 4$ . They ask which supplier deserves attention this week, which is a question about nodes rather than node-weeks, and the adequacy signal answers it. At each turn the eight nodes were ranked by hidden risk, and the set flagged above 0.10 was compared with the nodes that would go on to miss a milestone at its originally committed deadline. Four of the eight did.

**Table 7:** Nodes flagged by hidden risk against nodes that later missed a commitment

| Turns                                  | Precision     | Recall | False alarms | Reading                                           |
|----------------------------------------|---------------|--------|--------------|---------------------------------------------------|
| T0 to T5, before the crisis is visible | 1.00          | 0.75   | 0            | Three nodes flagged, all three later missed       |
| T6 to T18, crisis public               | 0.50 or below | 0.25   | 1 to 3       | The declared signal catches up and the gap closes |

For six consecutive turns, from the opening turn to the one before the foundry freeze, the signal flags exactly three nodes, all three of which go on to miss a commitment, with no false alarm (Table 7). The fourth is never flagged early, so recall stops at three of four. On every one of those six turns the worst-scoring node is the satellite prime, which misses its deadline at turn 8 and delivers at turn 14: the flag precedes the miss by eight turns, and it is raised while that node's own manager declares an urgency of 0.001. Figure 17 puts the whole campaign on one axis.



**Figure 17:** The whole campaign on one axis.

The flag does not come from a pessimistic declaration. It comes from an optimistic declaration against a computed state that is not. A system aggregating only indicators would have ranked that node by its indicators; a system collecting only declarations would have believed it. The gap between the two carries the information.

The repair of Section 7.5 sharpens the same flag without changing which nodes it selects. The satellite prime scores 26.6 out of 100 at turn 3 against 46.0 for the next worst node; after repair the same turn gives 2.2 against 44.6. Identical ordering, an order of magnitude more separation, which is what makes the flag readable on a dashboard.

One limit bounds all of it: eight nodes, four positive, one scenario. A precision of 1.00 over six turns on three flagged nodes is an encouraging observation, not an established rate.

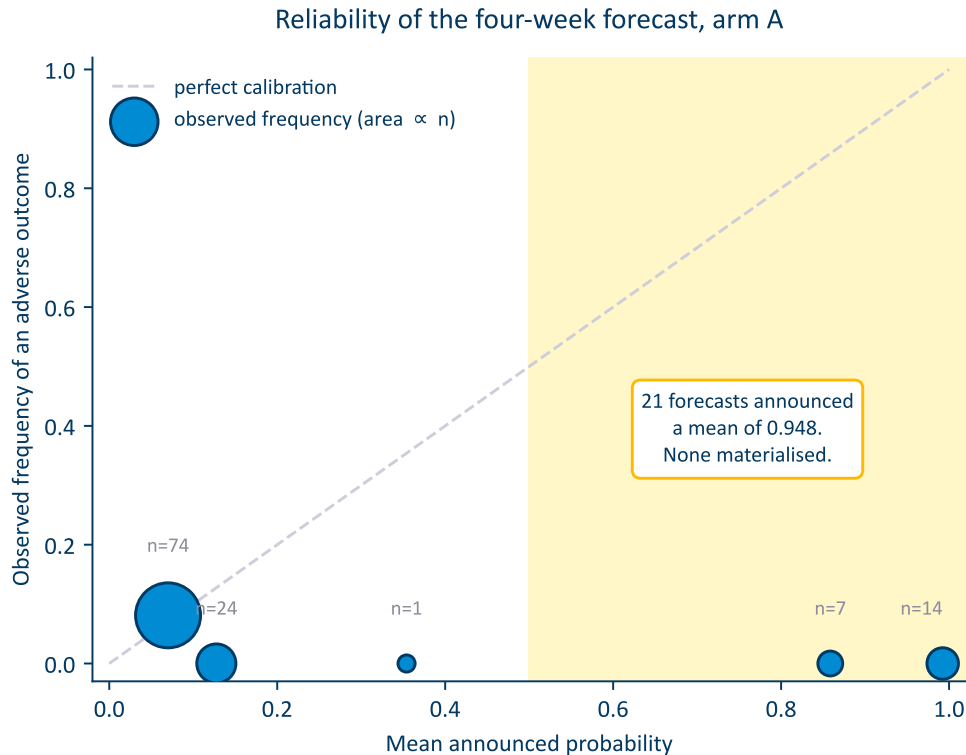
#### 7.4.2 The forecast layer, and the definition it is scored against

Turning that ranking into a probability attached to a named week is a further step, and measuring it made the rest of this section possible. Scored against the outcome definition the system itself applies, a missed milestone, an abandoned node or a critical event within four weeks, the first version of the forecast layer scored below the reference on both axes that matter, precisely enough to act on.

Discrimination is indistinguishable from chance: the area under the ROC curve ranges from 0.37 to 0.56 across horizons and every confidence interval contains 0.5. The positive class holds between two and six observations, too few to conclude either way.

Calibration is clearer and does not depend on that sample. The Brier skill against a forecaster announcing the base rate is between  $-3.4$  and  $-6.9$ , several times worse than the trivial baseline, and Figure 18 localises it.

Where the model announces a low probability it is right: 74 forecasts averaging 0.070 were followed by an adverse outcome 8.1% of the time. Where it announces a high probability it is wrong every time: twenty-one forecasts announcing a mean of 0.948 were followed by the predicted outcome zero times (Figure 18).



**Figure 18:** Reliability of the four-week forecast

The two need different repairs, which is Murphy's distinction (Murphy, 1973). A forecast that orders outcomes correctly but announces the wrong numbers is fixed by a thin post-hoc layer. One that does not order outcomes needs the model changed. Here the evidence for the second is weak and rests on few positives; the evidence for the first is strong and does not depend on that sample.

#### 7.4.3 Why: one block explains all of it

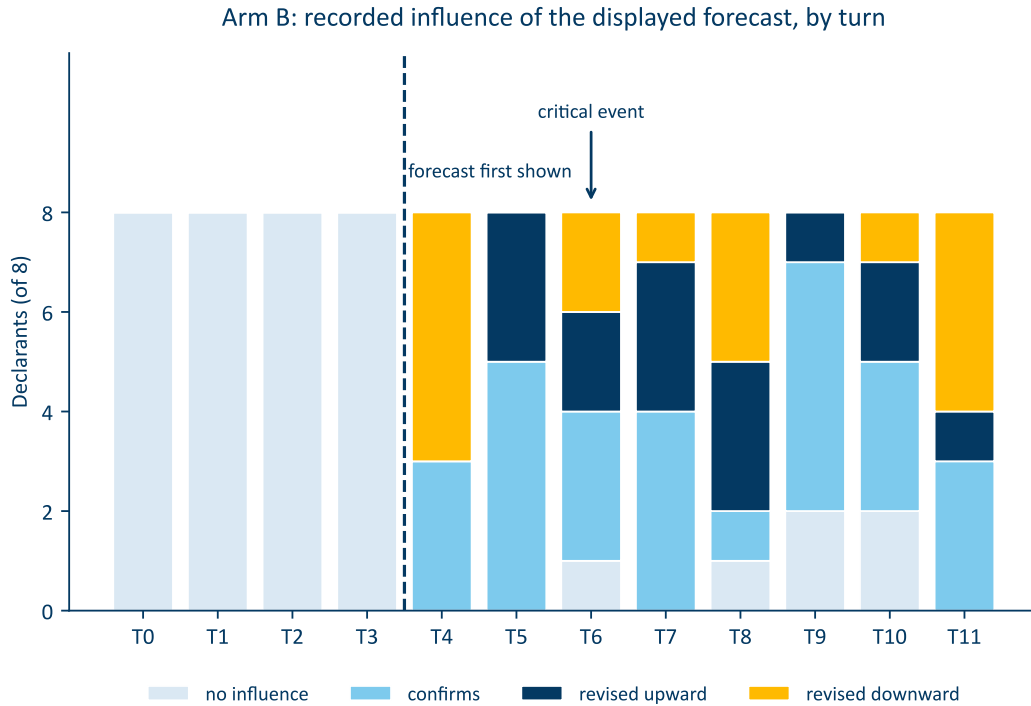
The explainability layer localised the cause. Across all twenty-one confident forecasts, the four-week probability and the milestone-miss probability never differ by more than 0.020, so the confident tail is the milestone channel and nothing else.

Reading the time block directly at the turn the scenario's critical event lands confirms the diagnosis from the other side: the block reads numerically zero at six of the eight nodes in the middle of a crisis. The cause is structural. The block computes milestone risk from a lead-time distribution against a deadline, while the event engine writes to capacity, cost and failure probability and never to any quantity on the time axis. The strongest form of the evidence is a count: across the whole campaign, eight nodes and forty-one indicator snapshots each, the accumulated-delay field the block would need was written a non-zero value exactly zero times. No path connects a shock to the quantity the block reads, so the block cannot respond to a disruption. It is blind by construction, and because it drives the entire confident tail, so is the forecast. Figure 21 draws the missing path and the one that replaced it.

This is a defect in the model rather than a property of the scenario. Any deployment would inherit it, and the repair is a change to the event mapping rather than to the mathematics. It is the single most actionable finding of the mission.

#### 7.4.4 What happens when the forecast is shown to the operator

For the first four turns the forecast is hidden and all eight declarants record no influence. At the turn it first becomes visible, all eight record one, and the direction is entirely one-sided: five revised their declared urgency downward and three had their existing assessment confirmed. Not one revised upward. The first upward revision anywhere appears the following turn, and the scenario's single critical event lands two turns after the moment every declarant was either reassured or confirmed.



**Figure 19:** Influence of the displayed forecast, by turn

Over the full campaign the balance is close to even, 15 upward against 16 downward (Figure 19), so the one-sided effect belongs to first contact rather than being a permanent bias.

Two things sharpen it. The forecast displayed comes from the same layer whose confident predictions never materialise and whose time block is blind to the shock about to arrive. And it appears as a numeric probability with a confidence interval, a format the automation-bias literature identifies as increasing reliance (Goddard et al., 2012; Bansal et al., 2021). The system presented an unfounded reassurance in its most persuasive available form, two turns before the event.

One qualification cuts against the strongest reading: the recorded field captures the effect on the declared level, not on the action taken, and the free-text notes show declarants still acting protectively. The effect is on the measurement, which is nonetheless the input the whole adequacy layer consumes.

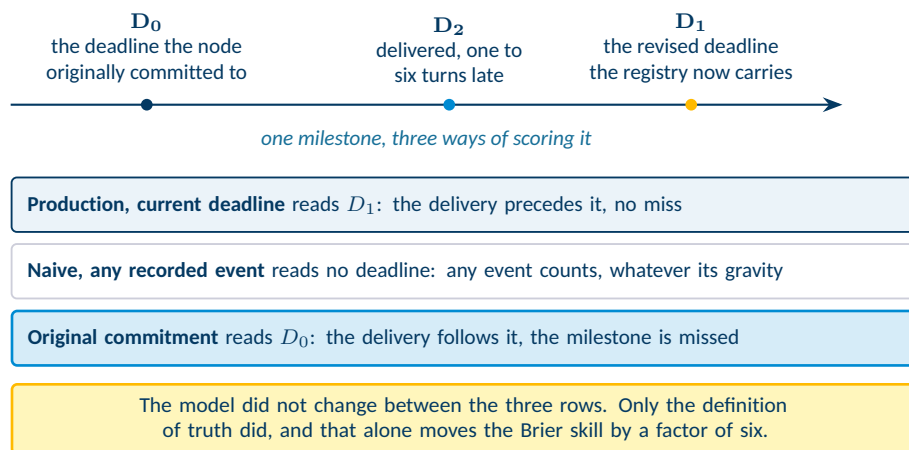
#### 7.4.5 The outcome definition moves the answer

A final measurement concerns how these results should be read at all. The identical one hundred and twenty forecasts were scored against three definitions of an adverse outcome, and the three disagree by more than any modelling choice made during the mission. Table 8 gives them at four weeks.

**Table 8:** The same forecasts under three definitions of truth, four-week horizon

| Outcome definition                            | $n$ | Base rate | AUC   | Brier skill |
|-----------------------------------------------|-----|-----------|-------|-------------|
| Production, current deadline                  | 120 | 0.050     | 0.485 | −3.36       |
| Naive, any recorded event                     | 75  | 0.387     | 0.484 | −0.64       |
| Original commitment, censored points excluded | 90  | 0.289     | 0.654 | −0.55       |

What separates them is which deadline they read, and Figure 20 follows one milestone through all three. The first is the definition the running system applies, and it is the one the measurement above is stated against. The second counts any recorded event regardless of gravity and ignores milestones entirely; it is the easy target, and choosing it would report a system five times better calibrated without a line of code being touched. The third keeps the production definition of an adverse outcome but reads a missed milestone against the deadline the node originally committed to, and drops the forecasts whose observation window runs past the end of the campaign.



**Figure 20:** The same milestone scored three ways.

The third row is the one the rest of this section uses, for a reason that is a matter of what a forecast is for rather than of statistics: a replanning is itself the admission of a slip, so a truth that lets a re-dating erase the miss scores the forecast against an outcome the forecast was supposed to warn about. A calibration figure quoted without its outcome definition therefore carries close to no information, which is a methodological result worth more to the programme than the score itself.

## 7.5 The Repair, and What It Measures

The diagnosis named one component and specified a repair, carried out and re-measured on the identical frozen campaign. Declarations and scenario are unchanged and the same one hundred and twenty forecasts are recomputed, so any difference belongs to the model.

### 7.5.1 Measurement corrected the recommended repair

The obvious reading is that the calibrated events should write a lead-time impact, and that is what the first version did. It does not work, for a reason that generalises. A completion date estimated as a full production cycle starting now scales the lead time by the fraction of work remaining, so anything routed into the lead time is scaled with it, and a shock evaporates precisely on the milestones closest to their deadline. On the node carrying the scenario's accident the four-week probability fell from

0.846 to 0.000 on a milestone that was in fact missed. Lost production time is additive: a four-week freeze costs four weeks whether the order is a tenth done or nine tenths done.

The completion model was rewritten accordingly, identically in the three independent estimators the system carries for the same quantity, the analytic time block, the Monte Carlo rollout and the PERT simulator:

$$\text{completion} = t + (1 - \text{progress}) \times \text{lead time} + \text{lost time}. \quad (2)$$

Five families of capacity-halting event now accumulate their duration into the node's lost-time indicator: machine failure, accident, strike weighted by the share of the workforce involved, material shortage weighted by criticality, and cyber incident. Loss of storage capacity deliberately does not, since it removes space rather than production hours. On the node carrying the scenario's accident, the first shock alone now writes 154.7 engine hours.

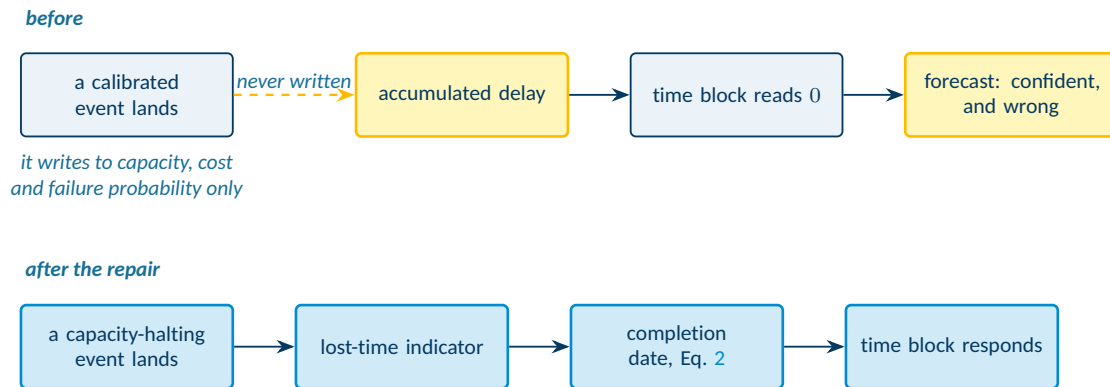


Figure 21: Before and after repair.

Repairing that exposed a second improvement. With the time block responsive, the milestone probability turned out almost binary: exactly zero or exactly one in 85% of cases, and two declarants had said so in their notes before it was measured. The rollout drew the lead time at random but took declared progress as exact, when it is a declared quantity like any other and should carry its own uncertainty. Progress is now drawn per replicate, with a standard deviation widened node by node in proportion to that node's own measured hidden risk, so the system uses its adequacy measurement to calibrate its own forecast.

## 7.5.2 What the re-measurement gives

Table 9 scores the recomputed forecasts against the original-commitment definition of Section 7.4.5, on the same censored samples of 112, 104, 96 and 90 observations.

Table 9: The frozen campaign rescored after repair

| Horizon | Brier skill |           |               | AUC    |           |               |
|---------|-------------|-----------|---------------|--------|-----------|---------------|
|         | before      | structure | + uncertainty | before | structure | + uncertainty |
| 1 week  | −1.278      | −0.646    | −0.687        | 0.736  | 0.793     | 0.729         |
| 2 weeks | −0.690      | −0.401    | −0.367        | 0.636  | 0.671     | 0.634         |
| 3 weeks | −0.654      | −0.360    | −0.286        | 0.641  | 0.657     | 0.602         |
| 4 weeks | −0.553      | −0.272    | −0.216        | 0.654  | 0.721     | 0.584         |

The structural repair improves both axes at every horizon: the calibration deficit is roughly halved and discrimination rises, with neither traded for the other. The progress-uncertainty term is a trade, improving calibration from two weeks onward and lowering the AUC; that last movement is not

interpretable, since with twenty-six positives the ninety-five percent interval on the four-week figure runs from 0.21 to 0.87. It was kept on the argument that a declared quantity should not enter a probabilistic forecast as a certainty.

Two things stand against it. The Brier skill is still negative at every horizon, and the milestone probability is still binary in 72% of cases, down from 85%.

**A third arm, in closed loop.** The repaired system was then run through a full nineteen-turn campaign in which the same eight declarants also chose actions from a fixed catalogue, applied to the chain and journalled: 152 declarations, no unresolved consistency failure. The headline figure flatters the tool and should not be used. No milestone was left unfinished in the closed loop against six missed in the frozen scenario, which sounds like six saved; read against the original commitments, five of eleven were met on their original date in each run. The loop changed which milestone was met, not how many, and with eleven milestones neither benefit nor cost is established.

That run also repaired the criticality ranking, which knocks out each node in turn and measures the damage at the final customer, and which collapses when many nodes saturate at once: every delta clips to zero, in exactly the crisis where it would be worth reading, on nine of the nineteen turns. A second sort key in log-survival,  $\ell(p) = -\ln(1 - p)$ , on a regularised twin of the same propagation, orders identically outside saturation and survives the clip inside it. Turns on which the deep suppliers can be ordered at all rise from 10 to 17 of 19.

Two behavioural observations carry. The turn-four disarming reproduced exactly, eight declarants of eight. And where the first campaign's foundry manager wrote that the tool was holding a low position through a widely reported crisis and took no action, the repaired run moved that node's four-week probability from 0.088 to 0.214 and the declarant replanned the milestone. The chain from a scripted shock to a displayed number to an action taken now closes.

## 7.6 A Second Chain, and the Domain It Establishes

A result measured on one scenario stays a property of that scenario until it is tested somewhere else. The last phase of the mission therefore built a second supply chain, chosen so that every design decision differs from the first, and ran the whole tool through it again. What came back establishes where the measurement applies.

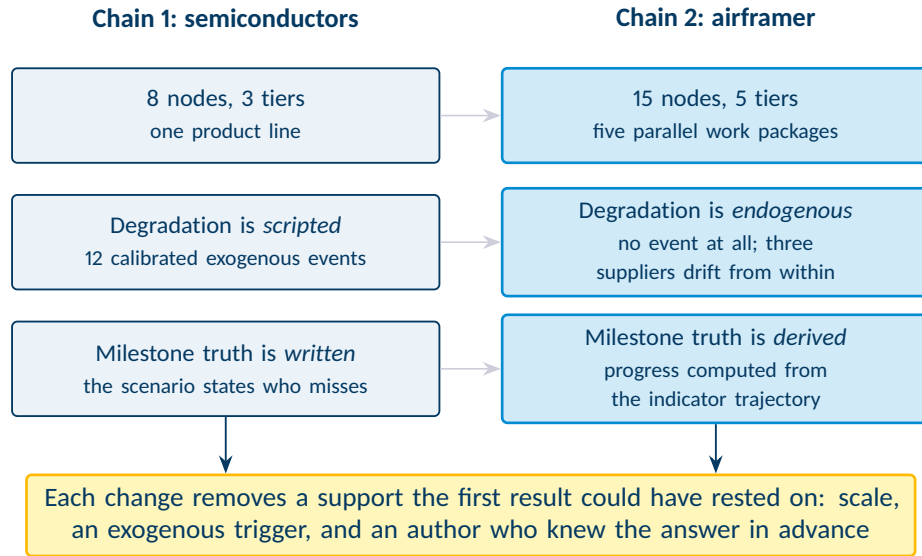
### 7.6.1 What was changed, and why each change is a real test

The second chain, AIRB, is an airframer's supply base: fifteen nodes over five tiers, from titanium and composite suppliers through structural, hydraulic and avionics subcontractors up to the final assembly line. Three properties separate it from the semiconductor campaign, and each removes a support the first result might have been resting on, so that whatever survives is a property of the tool.

Figure 22 sets the two side by side. The scale change is the mildest of the three. The second is sharper: AIRB carries no exogenous event whatsoever, so a supplier degrades only because its own indicators drift, and anything that was implicitly reacting to a scripted shock has nothing left to react to. The third is sharper still. Milestone truth on AIRB is computed from the indicators rather than written by the scenario author, through a health term that converts each week's lead time, equipment effectiveness and loss into a fraction of progress, accumulated until it reaches one:

$$\text{progress}_t = \min\left(\text{progress}_{t-1} + \frac{\text{health}_t}{d - s}, 1\right), \quad (3)$$

where  $s$  and  $d$  are the milestone's start and due turns. Whether a milestone is missed is therefore a consequence of the data rather than a decision, which removes the author from the answer. Twenty-two milestones are observable on the chain and five are missed, spread over three suppliers.



**Figure 22:** The two campaigns. The second chain was specified to differ on every axis that mattered, so that it would test the first result rather than reproduce it.

### 7.6.2 Where the signal applies, and where it does not

SupplyScore ran nineteen turns with fifteen isolated declarants, producing 285 declarations. The adequacy signal, which on the first chain flagged three future failures with no false alarm, does not select them here, and the reason is the most transferable result of the mission.

**Table 10:** Which ranking finds the three failing suppliers, second chain

| Ranking                     | Precision@3 | Turns | Reading                                  |
|-----------------------------|-------------|-------|------------------------------------------|
| Adequacy score              | 0.00        | 19    | Never selects a failing supplier         |
| Hidden risk $[U_r - U_d]^+$ | 0.00        | 19    | Never selects a failing supplier         |
| Forecast probability        | 0.56        | 15    | Selects nearly three times the base rate |
| Base rate                   | 0.20        | 19    | 3 failing suppliers of 15                |

The reason is visible in the first turn's declarations and is worth more than the result it explains. Table 11 gives the four highest and the three lowest declared urgencies at turn 0.

**Table 11:** Declared urgency at the opening turn, second chain

| Node                | Declared | Computed | Hidden risk | Outcome       |
|---------------------|----------|----------|-------------|---------------|
| Titanium supplier   | 0.982    | 0.655    | 0.000       | <i>misses</i> |
| Composite supplier  | 0.941    | 0.838    | 0.000       | <i>misses</i> |
| Machining           | 0.914    | 0.811    | 0.000       | <i>holds</i>  |
| Harness maker       | 0.876    | 0.542    | 0.000       | <i>misses</i> |
| Structures          | 0.535    | 0.990    | 0.456       | <i>holds</i>  |
| Systems integration | 0.496    | 0.952    | 0.457       | <i>holds</i>  |
| Final assembly      | 0.400    | 1.000    | 0.600       | <i>holds</i>  |

The three suppliers that go on to miss declare the first, second and fourth highest urgencies in the chain, from the opening turn, before anything has visibly gone wrong. A quantity defined as the shortfall of declared against computed cannot open on a node whose declaration is already near its

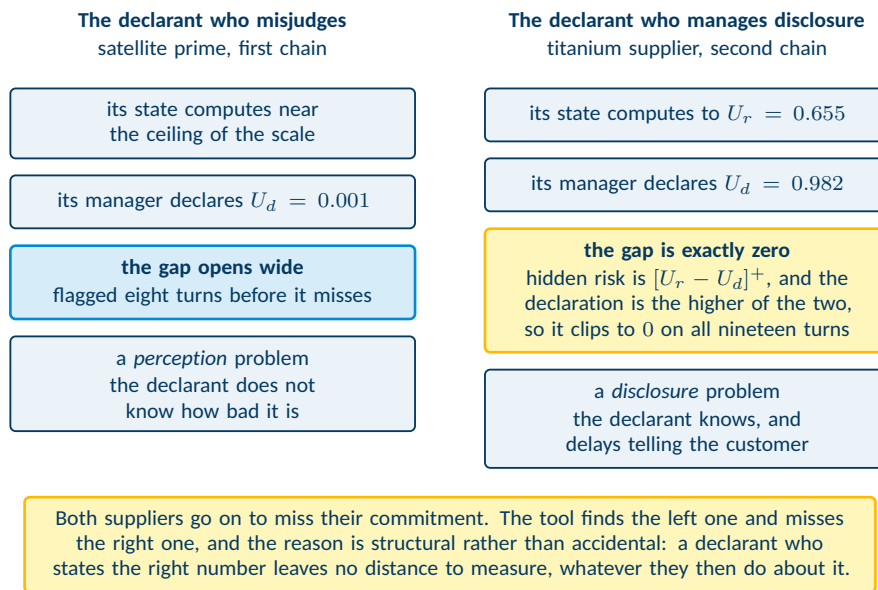
ceiling: their hidden risk is exactly zero for all nineteen turns, while the largest value the measure reaches anywhere on the campaign is 0.812, on nodes that hold.

What makes this diagnostic rather than merely negative is the free text submitted alongside those numbers. In the same forms, the three suppliers write that they intend to delay informing their customer until they have a recovery plan to present with the bad news. They minimise in action and maximise in measurement, and the two are not in conflict: a supplier who has understood its own situation perfectly well, and is managing the disclosure of it, declares high and acts late.

**What the tool can see, and what it cannot.** The adequacy gap is a distance between two numbers: what a node's state computes to, and what its manager declares. That distance only opens on a declarant who *gets their own state wrong*. It stays shut on a declarant who reads their state correctly, declares a number that matches it, and manages *when* the customer hears the bad news. There is no distance left to measure.

Two different problems have been separated here. *Misperception* is a measurement problem, and a questionnaire reaches it. *Non-disclosure* is a behavioural one, and no instrument that asks the supplier can reach it, because the supplier's answer is precisely the thing being managed.

Figure 23 sets one declarant of each kind side by side, with the figures each campaign recorded.



**Figure 23:** The boundary between two different declarants.

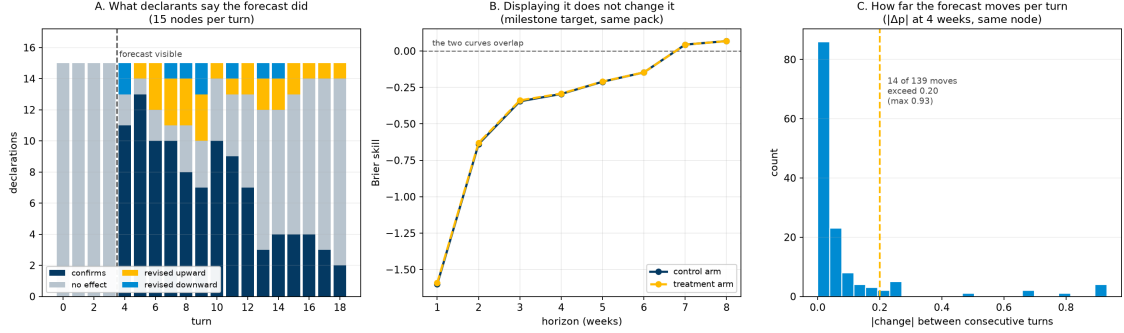
That distinction was not in the specification and was not in the state of the art, and it would not have appeared on a chain where the failing nodes were also the under-declaring ones. It tells a future user of SupplyScore which of their two problems it is for, which is the question they will actually need answered before deploying it.

### 7.6.3 Displaying the forecast, at five times the scale

The second chain was then replayed with the forecast displayed, on the same indicator pack and the same derived truth, so that display is the only difference between the two runs.

The first result is negative and simplifies the method for everything that follows. The forecast scores identically in both runs: a Brier skill of  $-1.601$  against  $-1.592$  at one week, an area under the ROC curve

of 0.826 in both, and an unchanged node ranking. Displaying a forecast does not corrupt it, because the forecast is computed from indicators the declaration channel never touches, so a treatment arm can be scored as a control arm. What the display changes is the declaration, and at fifteen declarants the pattern is clearer than at eight.

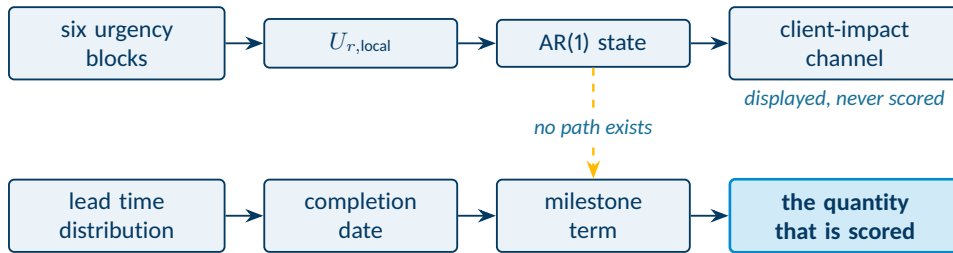


**Figure 24:** Second chain, run twice on one indicator pack. Left: what declarants report the forecast did to their judgement, by turn. Centre: the forecast scores identically whether or not it is shown, so the two curves overlap. Right: how far the four-week probability moves between consecutive turns on an unchanged node, which is what declarants cite when they stop reading it.

One reading is the one an engineer should act on. Reported influence falls from thirteen declarants of fifteen at first exposure to three at the final turn, and the declarants give a measurable reason: between consecutive turns, on the same node with the terrain unchanged, the four-week probability moves by more than 0.20 on roughly one node in ten, with a largest single move of 0.93. An advisory number that jumps like that teaches its reader to stop looking, and disengagement is the rational response rather than negligence.

#### 7.6.4 Does the forecast layer earn its complexity?

One question remained, and it is the one a reviewer asks last and an engineer should ask first. The forecast layer holds 0.56 precision where the adequacy gap holds 0.00, and it does so by running five hundred Monte Carlo trajectories per node per turn. Is that machinery doing the work, or is something much cheaper doing it? Reading the simulation loop gives the first answer, and Figure 25 draws it.



**Figure 25:** Inside the forecast rollout, the autoregressive state fitted to computed urgency feeds only the client-impact channel. The scored quantity depends on a lead-time draw and, on a chain with no exogenous event, on nothing else.

The consequence is testable, so it was tested rather than argued. A baseline was written to be as weak as possible: for a node whose next unfinished milestone falls due at turn  $D$  with fraction  $r$  of its work outstanding and nominal lead time  $L$  expressed in turns,

$$p(k) = \text{clip}\left(1 - \frac{D - t - k}{rL}, 0, 1\right). \quad (4)$$

No simulation, no fitted parameter, no calibration, and none of the six urgency blocks. The running service declines to forecast on 71 of the 225 node-turns, where no milestone is in flight; the baseline is silenced on exactly those, so both are scored on an identical sample at every horizon.

**Table 12:** Five hundred Monte Carlo trajectories against a two-number ratio

| Horizon | AUC        |          | PR-AUC     |          | Brier skill |          |
|---------|------------|----------|------------|----------|-------------|----------|
|         | simulation | baseline | simulation | baseline | simulation  | baseline |
| 1 week  | 0.826      | 0.908    | 0.153      | 0.396    | −1.601      | −4.851   |
| 2 weeks | 0.781      | 0.848    | 0.212      | 0.261    | −0.639      | −4.289   |
| 3 weeks | 0.757      | 0.791    | 0.252      | 0.258    | −0.345      | −3.949   |
| 4 weeks | 0.697      | 0.731    | 0.257      | 0.223    | −0.295      | −3.990   |
| 5 weeks | 0.651      | 0.676    | 0.275      | 0.221    | −0.211      | −3.798   |
| 6 weeks | 0.616      | 0.624    | 0.301      | 0.237    | −0.147      | −3.486   |
| 7 weeks | 0.667      | 0.583    | 0.438      | 0.239    | +0.044      | −3.180   |
| 8 weeks | 0.633      | 0.544    | 0.460      | 0.228    | +0.068      | −2.961   |

Table 12 gives the comparison, with the better of each pair in italics. The baseline ranks the misses better than five hundred trajectories at six horizons of eight, and by 0.082 of AUC at one week, which is the horizon a coordinator acts on. Since it reads the outstanding fraction from the same progress variable the truth is derived from, it was re-run in a form that reads nothing at all, taking outstanding work as the calendar fraction a perfectly healthy milestone would show. Its AUC and PR-AUC are identical to three decimal places at every horizon: the discrimination is entirely the ratio of weeks remaining to weeks required, and no part of it comes from the indicators.

The same table says what the simulation layer does contribute, and it is not discrimination. Its Brier skill is −1.601 at one week against the baseline's −4.851: the baseline produces an ordering and has no means of turning it into a probability, which is precisely what five hundred trajectories are for. It also holds its ranking at seven and eight weeks, at nearly double the PR-AUC, where a calendar ratio collapses below 0.60.

**What this establishes, and what it does not.** On this chain and this outcome, the forecast layer is a calibration wrapper around a lead-time ratio. That is a defensible thing for it to be, and at long horizons it is the better instrument. It is not what the interface presents, and it is not what the declarants of Section 7.6.3 believed they were deferring to.

Two limits bound the comparison. It holds for a chain carrying no exogenous event, where the event term of the scored quantity is empty by construction; on the semiconductor chain that term is populated and the argument does not transfer unchanged. And five positive observations at one week make an AUC difference of 0.082 an order of magnitude rather than an established gap. The test was optional, and was run because a layer that cannot beat a two-number ratio should not be shown to an operator as a probability.

## 7.7 Problems Encountered and How They Were Resolved

Three difficulties changed a technical decision. The two others, documentary access and the CPU-only budget, appear with their mitigation in the risk register of Section 5.5.

**The first model could not represent the problem.** The initial formulation expressed urgency as a single time-driven curve. It could not represent a node on schedule but out of capacity, which is a common and dangerous state. The resolution was to abandon it in April in favour of multi-block aggregation. The cost was roughly three weeks; the alternative was building a validated implementation of an inadequate model.

**Keeping the two signals independent had to be designed, not assumed.** A synthetic dry run written before the campaign derived the declared value from the computed one, which would have made any gap an artefact of the harness. It was discarded. The resolution was to state independence as a first-class design constraint: real urgency is fed from public time series of the crisis, declared urgency comes from participants who never see the computed value, and each result records which of the two could have reached the other. Running the declarants as language-model agents is what made that iteration affordable. A full nineteen-turn campaign replays in an afternoon, so a design flaw like this one was found and re-tested the same day rather than the following quarter.

**A test bench has to be able to contradict its author.** Each of the three tests of Sections 7.4 to 7.6.4 was designed while the outcome was still unknown, by someone with an obvious interest in one of the two answers. The resolution was procedural and carries to any R&D project: freeze the pass mark before the measurement exists, define the outcome in the same document as the threshold, and score any new instrument against the cheapest thing that could have replaced it.

## 7.8 A Parallel Workstream: Capacity Observed Rather Than Declared

The whole measurement rests on one hypothesis, and it is worth naming before the campaigns that use it: the weekly review is taken as a sincere report, so the indicators it carries and the urgency the same operator declares are two quantities rather than one. Putting at least one of those indicators beyond the declarant's reach altogether makes the hypothesis matter less.

A second system was built alongside the main work to do exactly that: it estimates the physical capacity of a warehouse from public satellite imagery, an observation no operator supplies and none can bias. Appendix D reports it in full. Two results carry. Counting loading docks from a nadir view is impossible, established by measurement rather than by opinion: an empty dock door under a canopy registers  $0.18\sigma$  of contrast against  $5.3\sigma$  for an occupied bay, and 8 cm imagery leaves it equally invisible. Restated as delimiting the equipped facade zone, the segmentation model reaches a global bias of  $-35\%$  on the derived dock count, which is directionally useful and not yet trustworthy. The two systems have not been connected, and no claim is made that they have.

# 8 Conclusion and Personal Assessment

## 8.1 Results Against the Objectives

Table 13 assesses each objective of Section 4.3 against its indicator.

**Quantitatively.** The delivered system covers the specified scope, carries an auditable write path and was validated at the intended scale. Two campaigns were run to completion across two supply chains, 23 nodes and 33 observable milestones, producing 510 recorded declarations.

The measurements themselves are numbers rather than impressions. On the first chain the adequacy signal flags three of four future failures with no false alarm over six consecutive turns. The forecast layer's Brier skill improves from  $-1.278$  to  $-0.687$  at one week and from  $-0.553$  to  $-0.216$  at four after repair, roughly halving the calibration deficit at every horizon while remaining negative. On the second chain the adequacy signal reaches precision 0.00 against a base rate of 0.20 while the forecast layer holds 0.56, and a one-line baseline outranks the simulation at six horizons of eight, by 0.082 of AUC at one week.

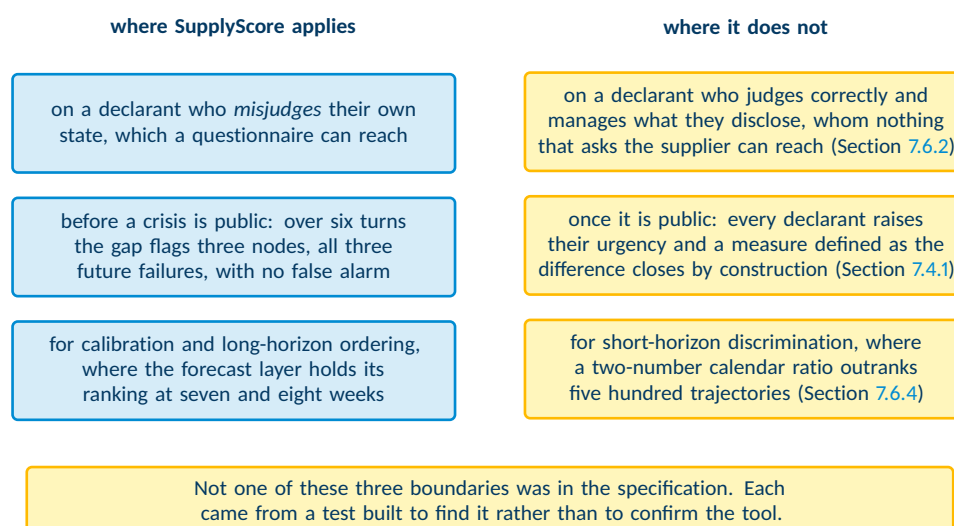
**Qualitatively.** The mission set out to determine whether the declared-versus-real gap can be measured well enough to act on. It can, at the level a coordinator actually works at, which is deciding where to look, and on a chain where the failing supplier is also the one misreading its own state. The

**Table 13: Objectives against outcomes**

|    | Status                           | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O1 | Met                              | Six-block aggregation with bidirectional propagation and an asymmetric adequacy score, each term exactly decomposable; aggregation choice justified against the compensatory alternative                                                                                                                                                                                                                                                       |
| O2 | Met                              | SupplyScore delivered: multi-tenant persistence, every write audited, operational and reloadable, with an automated test suite and a documented architecture                                                                                                                                                                                                                                                                                   |
| O3 | Met                              | Propagation and criticality services operational, validated by dedicated performance benchmarks at the scale of a real programme; the criticality ranking's collapse under network-wide saturation was found on the campaign and repaired (Section 7.5)                                                                                                                                                                                        |
| O4 | Met, with its domain established | At node level the signal is right on the first chain: over the six turns before the crisis is public it flags three nodes, all three of which later miss a commitment, with no false alarm (Section 7.4.1). As a calibrated probability it was scored, its limiting component located, repaired and re-measured (Section 7.5). The second chain and the baseline comparison then established where each layer applies (Sections 7.6 and 7.6.4) |
| O5 | Met                              | Scientific report delivered; every reported figure regenerated from raw artefacts by script rather than transcribed                                                                                                                                                                                                                                                                                                                            |
| O6 | Met, and reframed on the way     | A physical limit established by measurement rather than assumed: docks cannot be counted from a nadir view. The reframed segmentation model reaches a –35% bias on the derived count, which is directionally useful and scheduled for refinement in the remaining two months (Section 7.8)                                                                                                                                                     |

three tests that followed established where that claim holds, with a precision a favourable score alone could never have given it.

What the programme can now say about SupplyScore is a domain rather than a score, and Figure 26 states it. The tool is worth deploying where all three left-hand conditions hold, and not elsewhere.



**Figure 26:** The domain established for SupplyScore. A research and development programme is supposed to produce a tool *and* the evidence that says where it works; the right-hand column is that second half.

That is the outcome worth defending. The route from here is not a new idea but more chains, and Section 8.3 says which and why.

## 8.2 Limitations

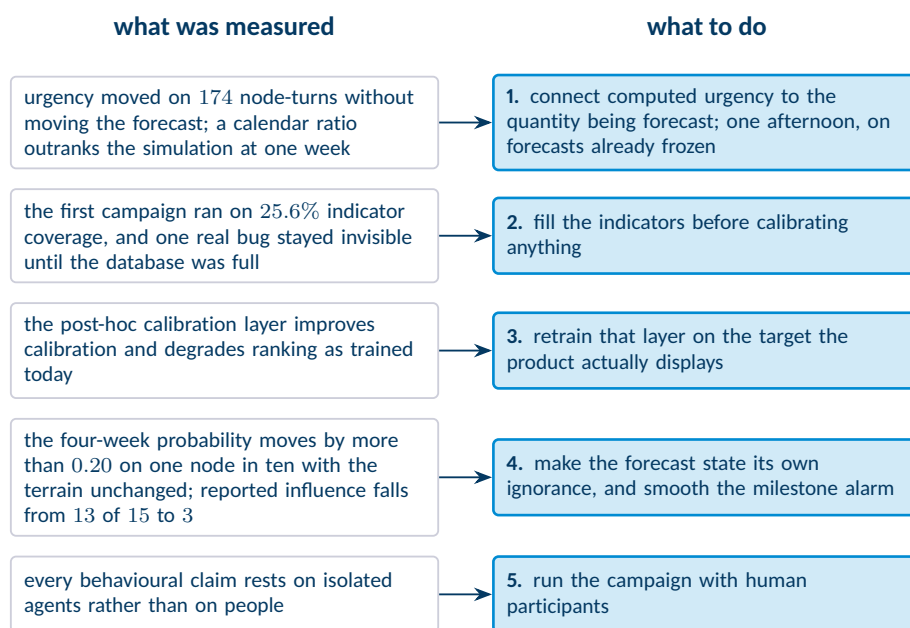
Four limits bound what this report supports, and three of them are limits of sample rather than of design. None calls the construction into question, and all three are answered by running the tool on more chains rather than by changing it.

The declarants were agents rather than human consultants, so every statement about declaration behaviour applies to that population. The positive class is small on both chains, four failures of eight nodes on the first and three of fifteen on the second, which is why the precision of 1.00 in Section 7.4.1 is reported as an encouraging observation rather than a rate, and why the AUC movements in Section 7.5 are stated as uninterpretable. And the carbon block remains the least exercised of the six for want of node-level data, the gap noted in Section 2.3.

The fourth is methodological and does not dissolve with more data. Every score depends on which of the three outcome definitions of Section 7.4.5 it is read against, and no number in this report should be quoted without one.

## 8.3 Perspectives

**For ALTEN.** The work leaves a system, two campaigns and an ordered list of what to do next. Figure 27 gives the list with, against each item, the measurement that produced it. Nothing on it comes from an opinion, which is why it can be ordered at all: the first item costs an afternoon and is measurable on forecasts already frozen, and the last is the only one that needs people.



**Figure 27:** Each line pairs a recorded measurement with the action it implies. The order is the order of cost: the first item is a one-line change scored on data that already exists, the last needs eight people for nine days.

**What the remaining two months do.** The three phases that close the internship are scheduled in Appendix B and each answers one of the limits above. New serious-game chains from 25 August to 19 September are the only action that moves a sample-bound figure. The vision workstream from 7 to 26 September attacks the  $-35\%$  bias of Section 7.8, where the measured driver of every gain so far has been annotated volume rather than architecture. The *Bilan Scientifique et Technique* then capitalises the internship for the programme.

One indicator class is missing rather than empty, and several declarants named it independently:

the forward booking horizon, meaning how far ahead a node's capacity is already sold. In the real semiconductor shortage that signal preceded the degradation of every flow indicator by months, and it is structurally invisible to the model as built. Any capacity-constrained link holds that information, so the gap is general rather than a semiconductor one.

**For the author.** The subject continues in the research thesis produced for Cranfield University. The question I would most like to pursue is the one the display experiment opened: what an advisory layer owes the person reading it when it does not know.

## 8.4 Personal Assessment

### 8.4.1 What I learned

Three technical capabilities grew over these months. *Supply chain modelling* came first, and it is the substance of Section 7.1: representing a multi-tier network, choosing which indicators carry rupture risk, and deciding how they combine. *Software engineering in Python* came with the application, built to be handed on rather than to run once. And *applied artificial intelligence* came from two directions at once, a probabilistic forecasting layer on one side and a computer vision segmentation model on the other.

The methodological lesson is the one I expect to reuse the longest: the independence of two signals has to be designed before either of them is. An early dry run derived the declared value from the computed one and had to be discarded, and the forecast layer took a declared progress figure as exact where it should have carried its uncertainty. Both times the symptom looked like a modelling defect and the cause was in the wiring of the measurement. From that came a habit that costs almost nothing: before trusting a new component, write the cheapest thing that could replace it and score them against each other, which is what produced Section 7.6.4.

With hindsight I would change two things. I would have scored the forecast layer against recorded outcomes far earlier, since doing so took a few hundred lines and located the defect immediately where months of implementation had not. And I would have written the outcome definition into the pre-registration alongside the thresholds.

### 8.4.2 Link with the ESTIA engineering competency framework

Table 14 maps the mission onto the competencies of the ESTIA framework that it genuinely exercised. Appendix C gives the detailed evidence.

### 8.4.3 What this mission clarified professionally

I came into this internship expecting the interesting difficulty to be in the mathematics. The mathematics behaved. The difficulty sits at the two joins: between a model and the world it describes, and between a model's output and the person meant to act on it. A digital twin is only as good as the understanding of the real operation behind it, and building that understanding is where most of these six months went.

That is what I now understand the research and development engineer's job to be, as distinct from the design engineer's or the consultant's: not only producing a system that works, but producing the evidence that says where it works, in a form the next person can check. The tests in Section 7 are worth more to ALTEN than the application they were run against, because the application is one programme's asset and the method transfers to any of them.

The mission confirmed the direction I want to take and made it more precise. I want technical work in simulation: turning the behaviour of the physical world into mathematical and computational

**Table 14: Competencies exercised during the mission**

|      | Competency                                                                    | Evidence from this mission                                                                                                                                    |
|------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CI1  | Analysis, synthesis, creativity, systemic and interdisciplinary approach      | Combining operations research, behavioural science, probabilistic modelling and software architecture into one measurement                                    |
| CI3  | Working in an international and intercultural context                         | Conducted and reported in English, under a double-degree programme, alongside an English-speaking colleague                                                   |
| CI5  | Identifying and solving complex or research problems using external resources | A state of the art built across published literature and the patent record, neither of which settled the question alone                                       |
| CE3  | Leading and controlling a project against objectives, deadlines and risks     | A four-touchpoint weekly cadence, ten phases, two documented replannings, and a risk register of which three items materialised                               |
| CE4  | Taking the socio-ecological transition into account                           | Carbon modelled as one of six first-class blocks, and the point where environmental data stops being available at decision granularity quantified             |
| CE5  | Researching, selecting, qualifying and diffusing information                  | A candidate dataset rejected on inspection; a registered appraisal of every retained source; the same work written for three audiences                        |
| CST1 | Defining an engineering strategy in a poorly known context                    | Converging a five-workstream assignment onto one defensible contribution, and abandoning the first model when it proved unable to represent the problem       |
| CST2 | Modelling and simulating a system, capitalising the knowledge produced        | The six-block model, its propagation, two campaign simulations, and a reproducible derivation of every reported figure                                        |
| CST7 | Building models, methods and tools to aid engineering, in an R&D perspective  | The application, the pre-registered protocol and the tests built against it, designed to be continued by the programme rather than to end with the internship |

models, at the boundary where those models meet the people who act on them. That is where the consequences are largest, and it is where an engineer can do most to shape how tomorrow's operations are run.

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## A Autorisation d'usage de l'IA par l'élève

The form of Annexe 7 was completed and signed by the company tutor on 25 August 2026. The use of generative AI tools is authorised, under the restriction written by hand on the form, *encadré par la convention ALTEN sur l'utilisation de l'IA*: within the scope of ALTEN's internal convention on the use of AI.

### Formulaire

|                       |                                            |
|-----------------------|--------------------------------------------|
| Nom de l'apprenant(e) | John HOARAU                                |
| Nom du tuteur         | Sébastien PERTHUISOT                       |
| Entreprise            | ALTEN Sud-Ouest, Direction de l'Innovation |
| Période concernée     | du 13 avril 2026 au 16 octobre 2026        |

### Rappel des risques pour l'entreprise en cas d'usage d'outils d'IA générative

L'utilisation d'outils d'IA par l'apprenti(e), sans encadrement ni précautions, peut exposer l'entreprise à plusieurs risques :

- divulgation involontaire de données confidentielles ou stratégiques lors de la saisie dans des outils en ligne ;
- perte de maîtrise sur la propriété intellectuelle des contenus générés (textes, codes, visuels) pouvant être réutilisés par la plateforme ;
- dépendance à des outils tiers dont les résultats peuvent être imprécis, biaisés ou juridiquement discutables ;
- non-conformité au RGPD si des données à caractère personnel sont traitées par des services non conformes.

### Autorisation du tuteur

☒ J'autorise l'élève à utiliser des outils d'intelligence artificielle générative, en pleine connaissance des risques mentionnés ci-dessus.

☐ Je n'autorise pas l'usage d'outils d'IA générative pendant la période en entreprise.

Commentaires ou restrictions spécifiques éventuelles :

*encadré par la convention ALTEN sur l'utilisation de l'IA.*

Fait à *Talence* le *25/08/26*

Signature du tuteur / maître de stage :



Figure 28: The authorisation form, signed.

## Declared use of AI tools in this mission

For transparency, and independently of the authorisation above, the use made of generative AI during this internship is declared here in full.

Large language models were used in two distinct capacities, which should not be confused.

First, as an *object of study*. The eight declarants of the experimental campaign reported in Section 7.3 are language-model agents. This is an experimental design choice, stated as such throughout the report, and its consequences for the validity of the results are discussed rather than glossed over.

Second, as a *writing aid*, for language correction and document formatting. Every scientific claim, every numerical result and every interpretation in this report is my own, and I am able to justify and explain each of them. Every figure was produced either by the project's own code or by hand. No confidential ALTEN data was submitted to any external service in the course of that assistance.

## B Detailed Planning

This appendix expands the schedule of Section 5.4 into the weekly rhythm that produced it, and records the two planning revisions made during the mission.

### Weekly cadence

Table 15 lists the standing meetings that structured every week of the mission.

Table 15: Standing weekly meetings

| Day       | Meeting                               | With                    | Purpose                              |
|-----------|---------------------------------------|-------------------------|--------------------------------------|
| Monday    | Planning, Green Supply Chain activity | É. Lardy                | Set the objective for the week       |
| Tuesday   | Scrum                                 | É. Lardy                | Progress, blockages                  |
| Wednesday | Scrum                                 | É. Lardy                | Progress, blockages                  |
| Wednesday | Supply chain regular point            | S. Perthuisot           | Scientific direction and arbitration |
| Thursday  | Scrum                                 | É. Lardy                | Progress, blockages                  |
| Friday    | Demonstration                         | É. Lardy, S. Perthuisot | Show working output, not describe it |

Six standing contact points a week is a demanding cadence for a research placement. Its effect was to bound how long any misconception could survive: the Friday demonstration in particular required something to be shown rather than reported, which repeatedly exposed gaps between what was believed to work and what did.

### Planning revisions

**June: validation over extension.** The original plan allocated the second half of the mission to widening the model's scope. It was reallocated to validating what already existed. The argument accepted at the regular point was that extending an unvalidated model compounds risk rather than reducing it. This revision is why the mission produced a measured result rather than a larger unmeasured system.

**July: substitution of the campaign population.** The campaign was designed for eight human consultants across nine working days. When that could not be secured within the mission window, the protocol was executed with isolated agents instead. The record format was deliberately identical, so the substitution changed the population without changing the instrument. Every behavioural

conclusion was weakened accordingly in the reporting rather than being stated at the strength the original design would have supported.

## Comparison of planned against actual

Table 16 compares the time allowed for each phase with the time it actually took.

**Table 16:** Planned against actual, by phase

| Phase                            | Planned          | Actual         | Comment                                                                                       |
|----------------------------------|------------------|----------------|-----------------------------------------------------------------------------------------------|
| Framing                          | 3 weeks          | 2.5 weeks      | On plan                                                                                       |
| First urgency model and pipeline | 6 weeks          | 5.5 weeks      | Stopped early once judged inadequate                                                          |
| Capacity from imagery            | 4 weeks          | 8 weeks        | Ran in parallel; the original objective proved unreachable and the problem had to be reframed |
| SupplyScore build                | 4 weeks          | 2 weeks        | Interfaces already established by the abandoned pipeline                                      |
| Experimental design              | 2 weeks          | 3 weeks        | Freezing the protocol properly took longer than allowed                                       |
| First campaign and diagnosis     | 3 weeks          | 3 weeks        | On plan                                                                                       |
| Independent validation           | <i>unplanned</i> | 3.5 weeks      | Added after the diagnosis: repair, second chain, baseline comparison                          |
| Calibration campaigns            | 4 weeks          | <i>planned</i> | 25 August to 19 September                                                                     |
| Vision model                     | 3 weeks          | <i>planned</i> | 7 to 26 September, overlapping the above                                                      |
| Scientific report and defence    | 3 weeks          | <i>planned</i> | 28 September to 16 October                                                                    |

The overall schedule held. Two overruns had to be absorbed. The imagery workstream took twice its allowance because the original objective turned out to be physically unreachable and the problem had to be reframed before anything could be built; the time came out of the SupplyScore build, which the abandoned pipeline had already de-risked. The validation phase was unplanned entirely and consumed three and a half weeks, released by the framing and build phases finishing early. The one systematic underestimate was the effort required to freeze an experimental protocol to a standard where the hypotheses could not later be adjusted, which took half again the time allowed and was worth every day.

## What the remaining two months carry

The mission runs to 16 October and this report is submitted on 24 August. The remaining time carries no new construction, and it is loaded to roughly the same weekly effort as the phases already closed.

1. *Calibration campaigns, 25 August to 19 September.* New serious-game chains, built the way the airframer chain was, run to widen the evidence base. This is the single action that moves every sample-bound figure in Section 8.2, and it is also what allows the forecast layer to be recalibrated on more than one scenario.
2. *Vision model, 7 to 26 September, overlapping the above.* The facade segmentation model of Section 7.8 carries a global bias of  $-35\%$  on the derived dock count. Annotated volume, not architecture, drove every gain measured so far, so the work is data collection and re-training rather than redesign.
3. *Scientific report and defence, 28 September to 16 October.* The *Bilan Scientifique et Technique* that capitalises the whole internship for the programme, and the defence.

Any rate quoted in this report may therefore move. The constructions and the diagnoses will not.

## C Detailed Competency Evidence

This appendix gives the evidence behind Table 14, competency by competency, so that each claim can be checked against something in the record rather than taken on assertion.

**CI1 – Analysis, synthesis, curiosity, creativity and innovation, approaching a problem systemically and across disciplines.** The measurement built during this mission required combining four fields that do not normally cite one another: deadline-driven value from operations research, the mere-urgency effect from behavioural science, probabilistic aggregation, and the software architecture needed to make the result auditable. The contribution claimed is not any one of them but the combination, which Section 6.8 shows to be absent from the reviewed literature.

**CI3 – Working in an international and intercultural context.** The mission was conducted, documented and defended in English, within a double-degree programme with a British university. A colleague on the same research programme worked in English on the adjacent representation problem, and the weekly demonstration was the point at which the two strands had to be made intelligible to each other.

**CI5 – Identifying and solving complex or research problems, mobilising scientific and technical knowledge and external resources.** The subject was a research problem with no established answer, and settling that it had none required two distinct bodies of evidence: the published literature, which says how to compute risk and how to elicit a judgment, and the patent record, which says what industry has protected. Neither alone would have closed the question, since a working technique can be commercialised without publication and a published one can go unclaimed (Section 6.6).

**CE3 – Leading, steering and controlling a project to reach its objectives, in cost, deadline, performance and risk.** Seven phases with dated milestones; a weekly cadence of six standing meetings; a documented mid-mission replanning that traded scope for validation; and a risk register of four items of which three materialised and were handled. An unplanned validation phase was absorbed without moving the end date, using the margin released by two phases that finished early (Appendix B). Cost is the one dimension not exercised, since no budget was managed.

**CE4 – Taking the socio-ecological transition into account.** Carbon is one of the six blocks of the model, ranked alongside cost and delay rather than applied as a correction, and the programme itself falls within the share of ALTEN's R&D devoted to environmental innovation (Section 2.3). The honest qualification, stated in Section 8.2, is that this block turned out to be the least exercised of the six for want of data.

**CE5 – Researching, selecting, qualifying and diffusing information, adapting it to its context.** The qualifying step is the one worth evidencing: a candidate dataset was inspected before use and rejected on three measurable signatures of synthetic data, and the node it would have fed was driven by calibrated events instead. On diffusion, the same work was written three ways for three audiences: a scientific report for the company, a research thesis for the partner university, and this mission report.

**CST1 – Defining an engineering strategy by choosing methods and tools suited to a known or poorly known context.** The assignment as signed covered five workstreams. Converging it onto one defensible contribution, and recording the boundaries explicitly rather than leaving them implicit, was the first engineering decision of the mission. The second was abandoning the initial urgency formulation once it proved unable to represent a node on schedule but out of capacity.

## **CST2 – Modelling and simulating a system, and capitalising the information and knowledge produced.**

The six-block model with its bidirectional propagation; two campaigns, one replaying a documented historical crisis and one built with milestone truth derived from the indicators rather than scripted (Equation 3); and a derivation pipeline that regenerates every reported figure from raw artefacts so that no number in any of the three deliverables is transcribed by hand.

**CST7 – Designing and building models, methods and tools to aid engineering and improve business processes, including in a research and development perspective.** The application and the experimental protocol were both built to be continued rather than to end with the internship: a documented architecture, an automated test suite, a frozen and hashed protocol, and a written record of the next steps in order.

The research contribution proper is the test bench as much as the tool: a scoring against recorded outcomes, a second supply chain built on different assumptions, and a minimal baseline. All three transfer to any model this programme builds next, and are documented as a method rather than as an episode of this mission.

## **D Capacity Observed Rather Than Declared**

The whole measurement rests on the hypothesis stated in Section 7.3: that the weekly review is a sincere report, so the indicators it carries and the urgency the same operator declares are two quantities rather than one. The first campaign made it worth putting at least one of those indicators beyond the declarant's reach altogether.

A second system was built alongside the main work to do exactly that: it estimates the physical capacity of a warehouse from public satellite imagery, an observation no operator supplies and none can bias. The two have not been connected, and no claim is made that they have.

### **A physical limit, established by measurement**

The original objective was to count loading docks, and it cannot be done from a nadir view. An empty dock door under a canopy produces no signal from directly overhead: measured contrast at verified empty bays is  $0.18\sigma$ ,  $0.22\sigma$  and  $0.01\sigma$  across a full sweep of assumed depths, while occupied bays on the same facades register between  $5.3\sigma$  and  $6.4\sigma$ . The instrument is sensitive enough; the signal is absent. On 8 cm imagery the doors stay equally invisible, which locates the cause in occlusion by the canopy roof rather than in resolution: a gain of  $2.5\times$  in ground sample distance recovers no additional door.

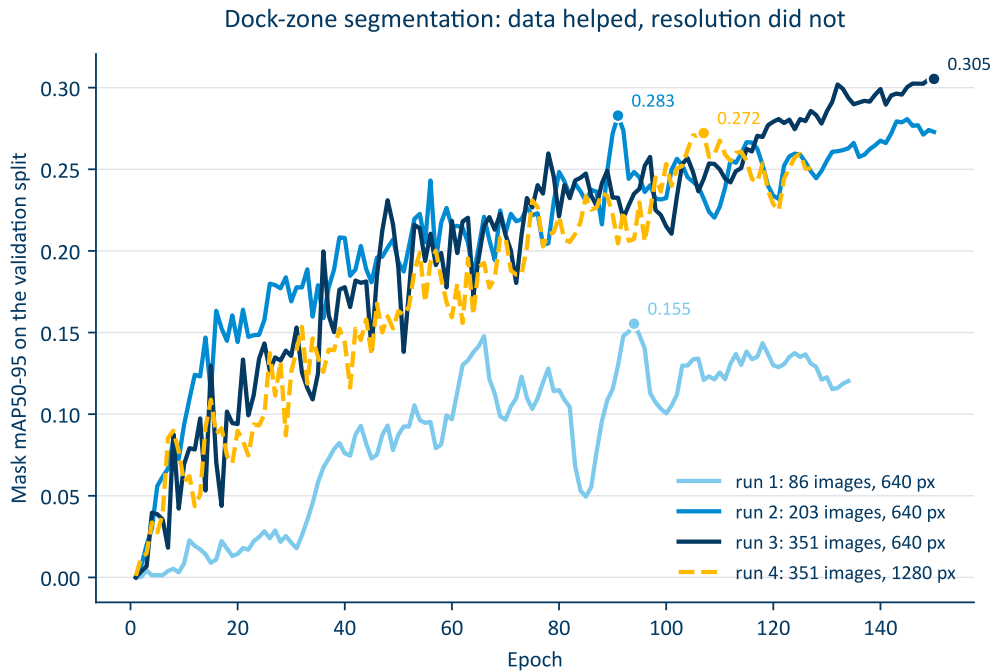
Any method claiming to count empty doors one by one from overhead is therefore recovering an artefact. Reaching that conclusion cost three weeks and saved the workstream from optimising a quantity that is not in the image.

### **The reframing, and what actually moved the model**

The problem was restated as one the imagery can answer: delimit the equipped facade zone, and derive the dock count from its length against a measured pitch. A segmentation model was trained on hand-annotated imagery across four runs (Figure 29).

Annotated volume drove the gain: quadrupling the training set from 86 to 351 images took mask mAP50-95 from 0.155 to 0.305. Resolution did not. Run 4 is identical to run 3 except for a doubled input size and is worse on every metric, because at a fixed data budget on a CPU-only allowance the extra pixels cost more than they bought. The remaining work is therefore data collection, not architecture search.

What is scored also had to change. Detection metrics measure shape overlap, but capacity depends only on the length of the equipped zone, and a polygon drawn too deep still yields the correct count. The project scores itself on the error in the derived count rather than on mAP. On the reference run



**Figure 29:** Validation mask mAP50-95 across the four training runs. Quadrupling the annotated set roughly doubled the score; doubling the input resolution at a fixed data budget made every metric worse.

that gives a global bias of  $-35\%$ , a median error of  $62\%$  on images containing docks, and  $89\%$  of true negatives correctly left empty: directionally useful, not yet trustworthy, which is why it appears in Section 8.3 rather than in the delivered scope.

### An annotation supplier that reported its own work as verified

An automated third-party annotation process returned 442 annotations and reported its own output as verified and complete. It was unusable: it had placed a dock zone along every edge of every building without examining the imagery, then subdivided each zone at the nominal pitch, giving an implausible  $88\%$  positive rate. All of it was discarded and the dataset re-annotated by hand; an audit of the manual work found 11 problems across 687 zones, a defect rate of  $1.6\%$ .

Checking cost one afternoon. Not checking would have cost a model trained to reproduce a fiction, and every downstream number with it. The same discipline rejected a synthetic logistics dataset in Section 5.3 and produced the baseline comparison of Section 7.6.4.